4. Introduction to ArcGIS

Objectives

1. You should be able to navigate with ArcGIS interface and component tools

What is ArcGIS?

- ArcGIS is a family of software products from GIS Industry leader esri Understanding our world.

ArcGIS

ArcGIS is a complete system for designing and managing solutions through the application of geographic knowledge.

- ArcGIS for Desktop
- ArcGIS for Mobile
- ArcGIS for Server
- ArcGIS Online

What's New in ArcGIS 10 »

Data

Esri data provides a range of ready-to-use, high-quality data for your GIS visualization and analysis projects.

- Basemaps
- Imagery Data
- Demographic, Consumer, and Business Data
- Free Data

Developer Tools

Developer tools allow you to add mapping to your web, mobile, and desktop applications using a choice of languages.

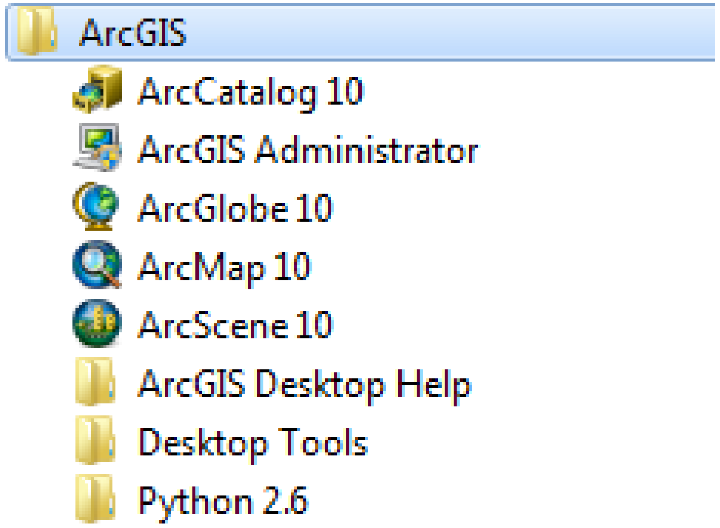
- Esri Developer Network
- Mobile Runtime SDKs
- Web APIs

What is ArcGIS?

- We will be using ArcGIS Desktop
- Three licensing levels:
  - Basic/ArcView – basic desktop package, $
  - Standard/ArcEditor – adds comprehensive editing, $$
  - Advanced/ArcInfo – adds advanced spatial analysis and high-end cartography, $$$
- We are using ArcInfo

What is ArcGIS?

- ArcGIS Desktop application suite:
  - ArcMap
  - ArcCatalog
  - ArcScene
  - ArcGlobe
- Other components include
  - ArcToolbox and ModelBuilder
  - ArcGIS Administrator

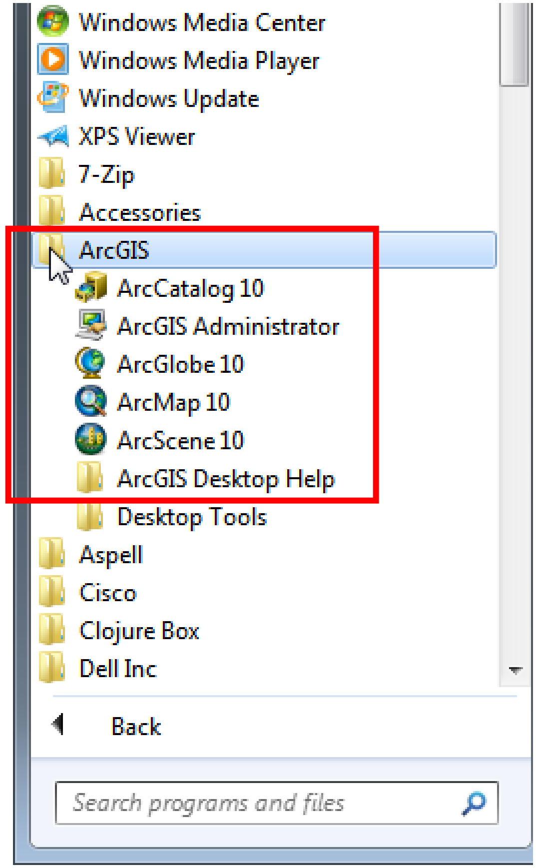


What is ArcGIS?

- How do you access these?
- From the start menu

  >> All Programs

  >> ArcGIS

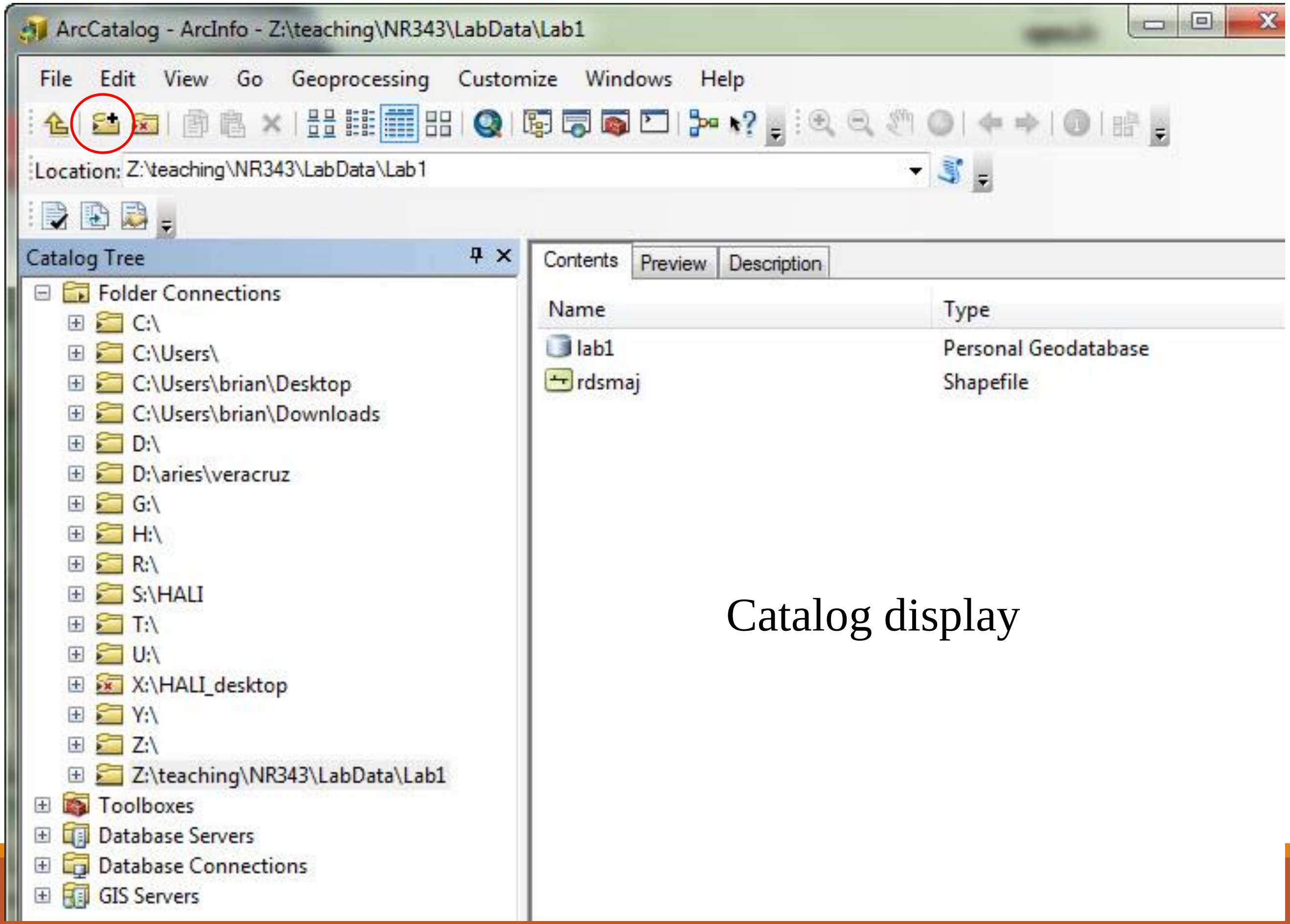


ArcCatalog

- for organizing and managing spatial and tabular data

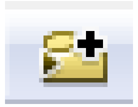
Main menu
Standard Toolbar

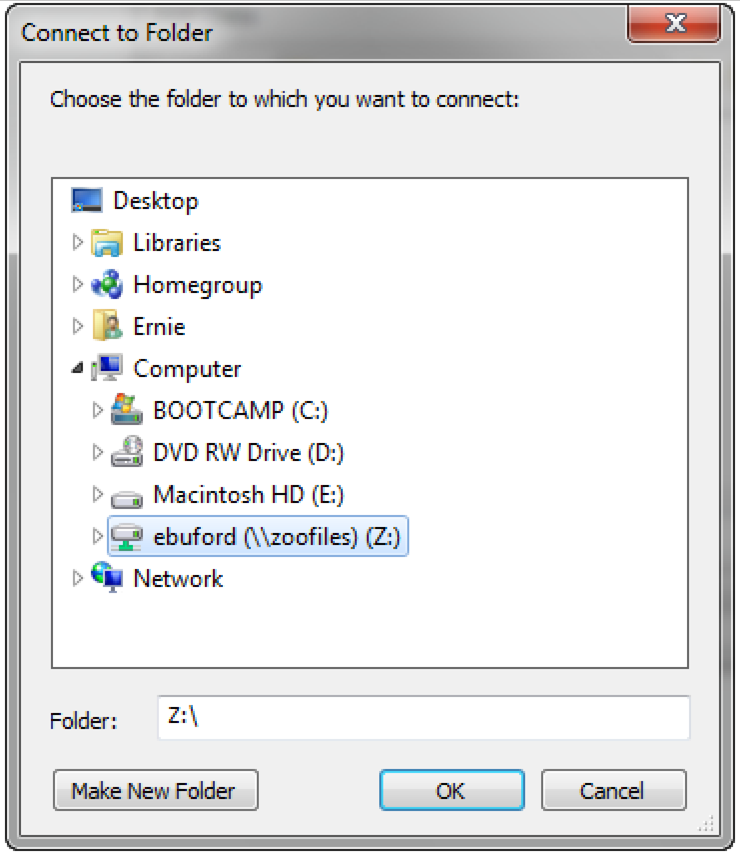
Catalog tree



ArcCatalog

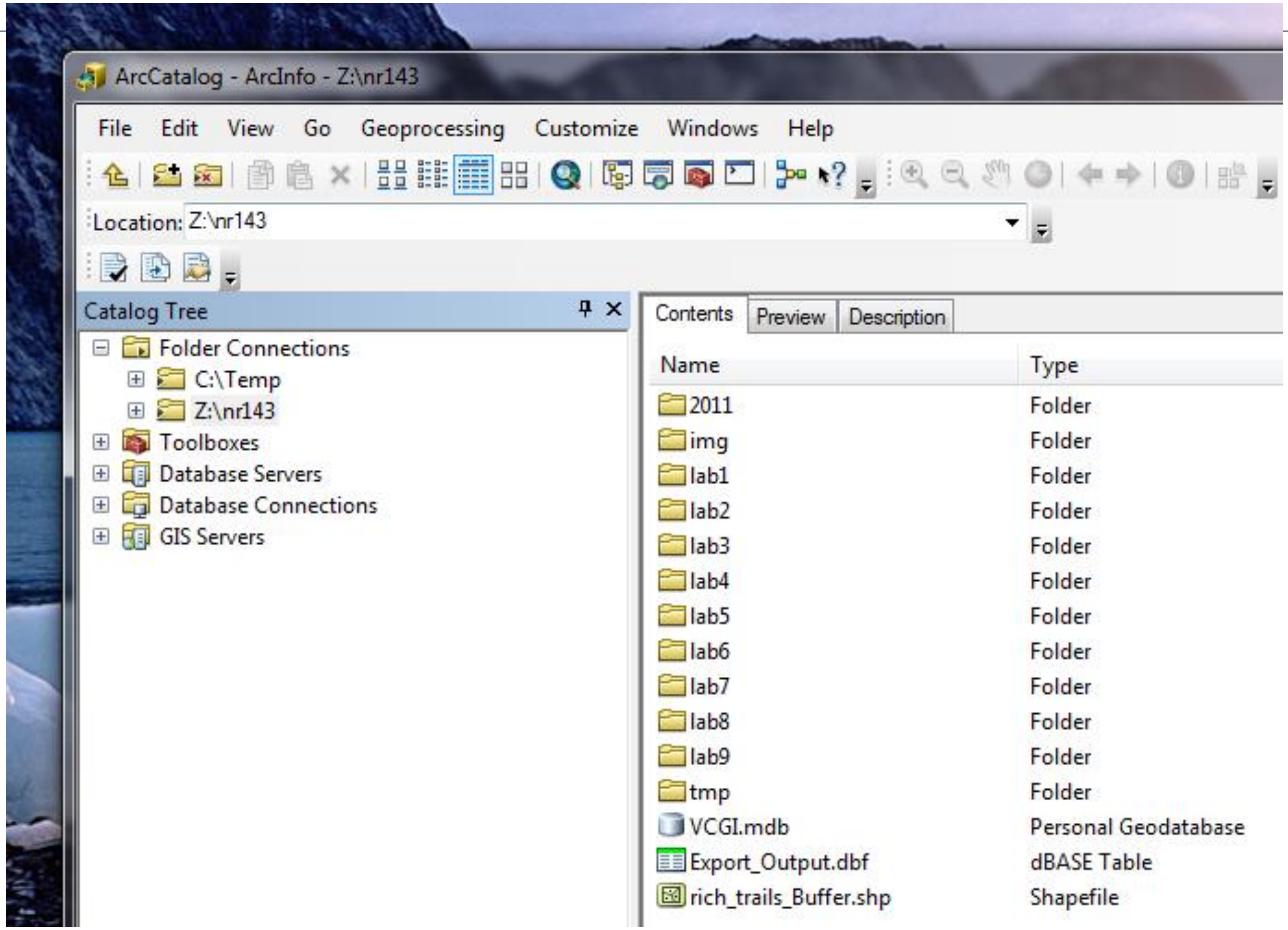
- Once a network share is mapped as a drive letter, you can create a connection to it or one of its subfolders





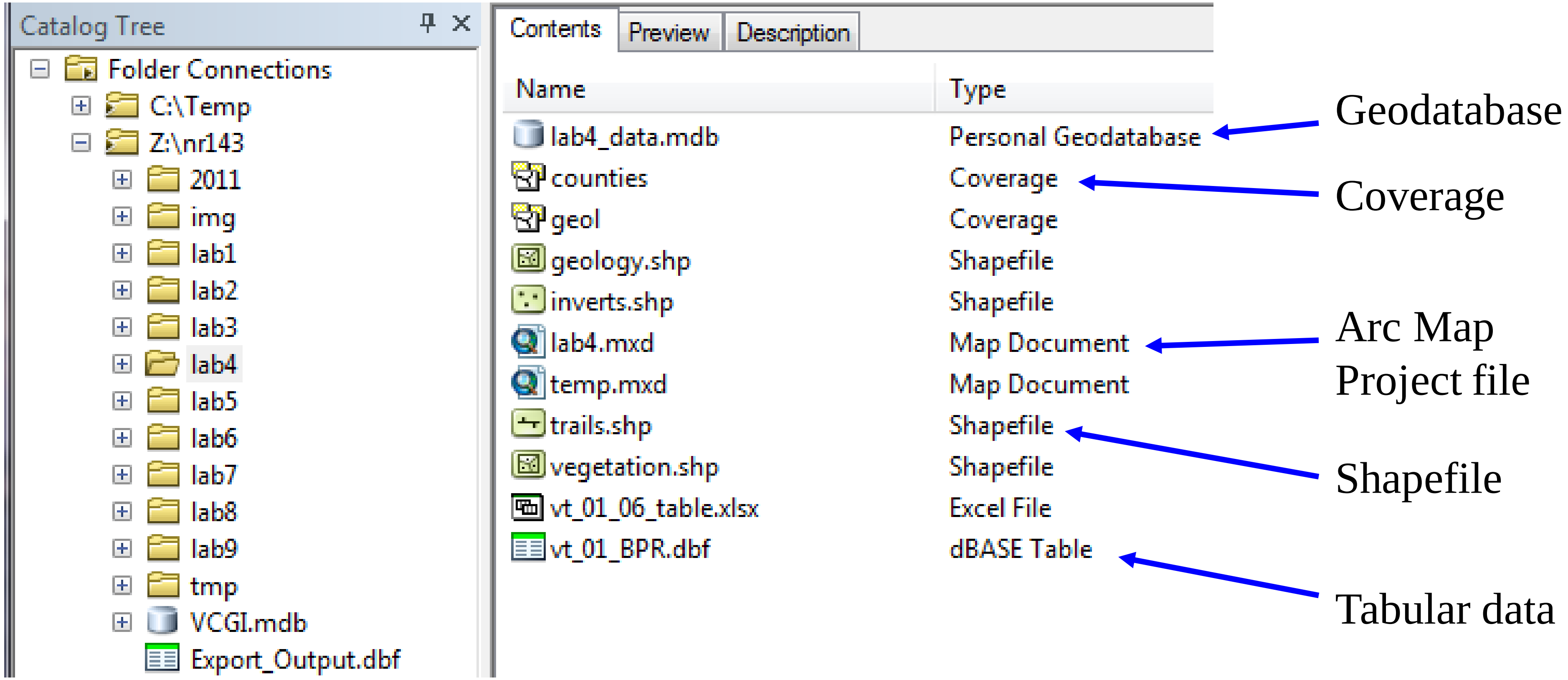
ArcCatalog

- For instance, here are the contents of the nr143 folder on my Zoo account, which I made a folder connection to:



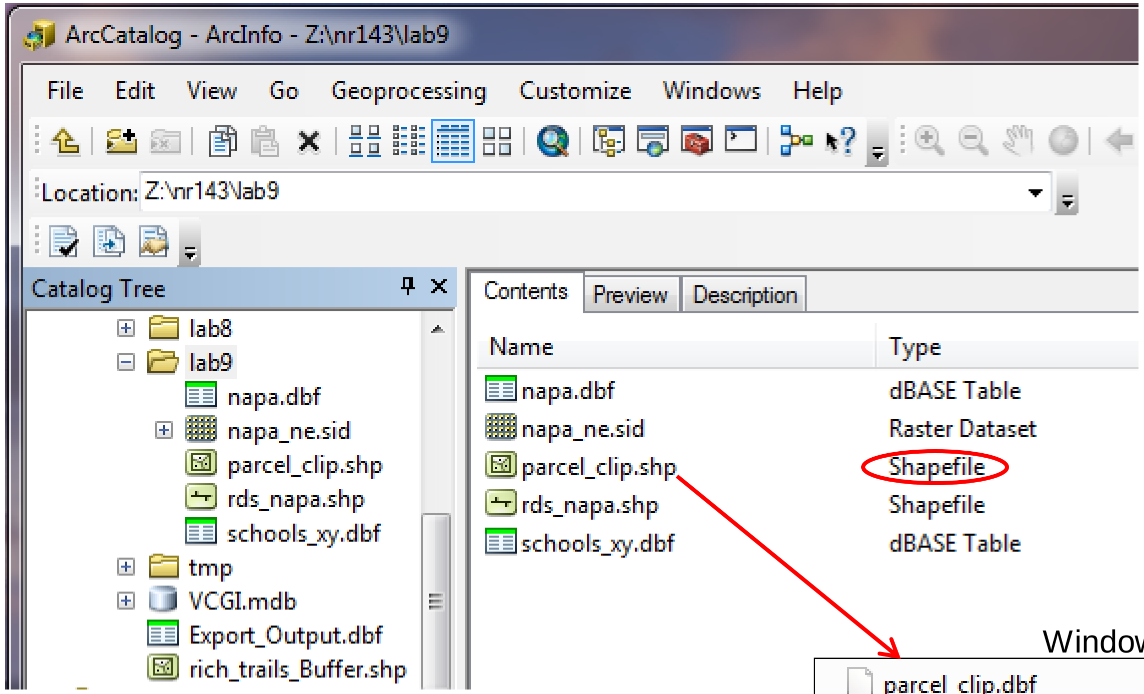
ArcCatalog

- There are five types of data shown here

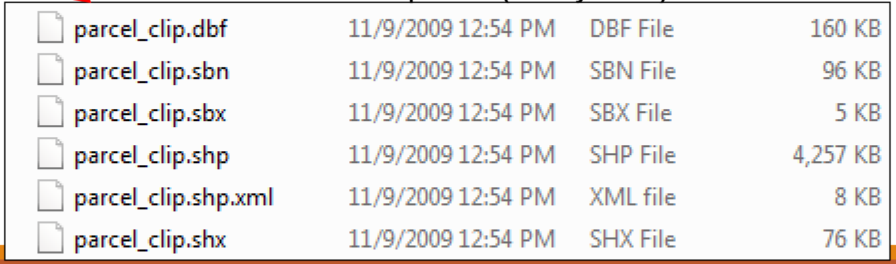


Data types

- **Shapefile**: native file format for ArcView 3.x



Windows Explorer (file system) view:

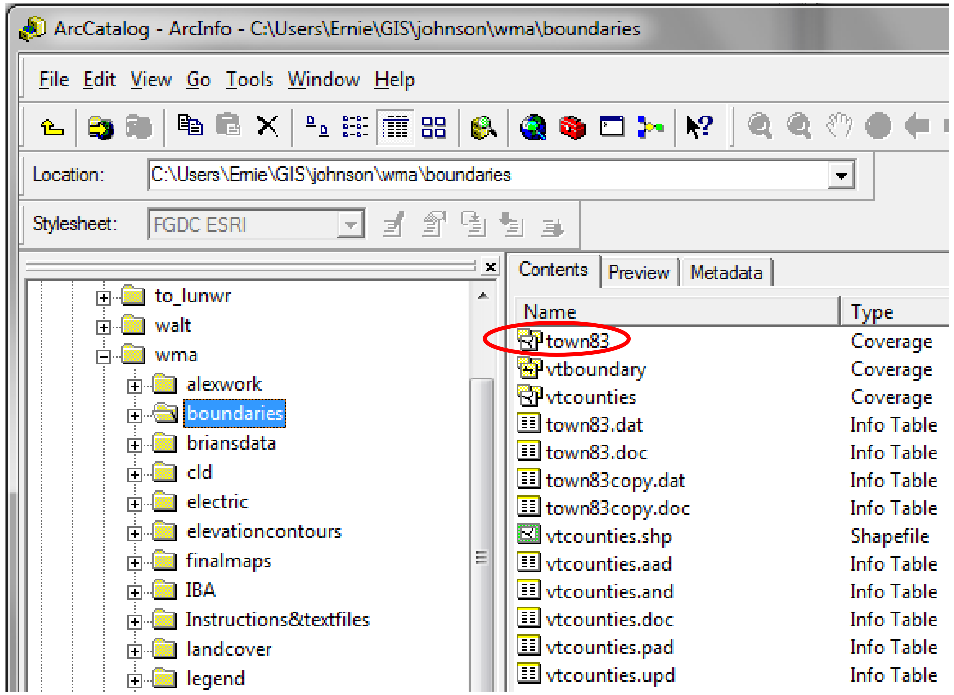


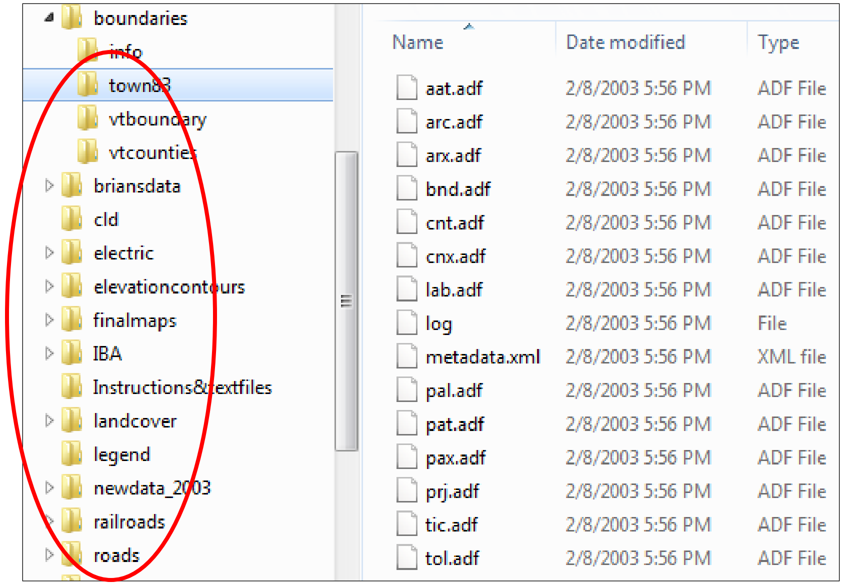
One spatial data layer, several files

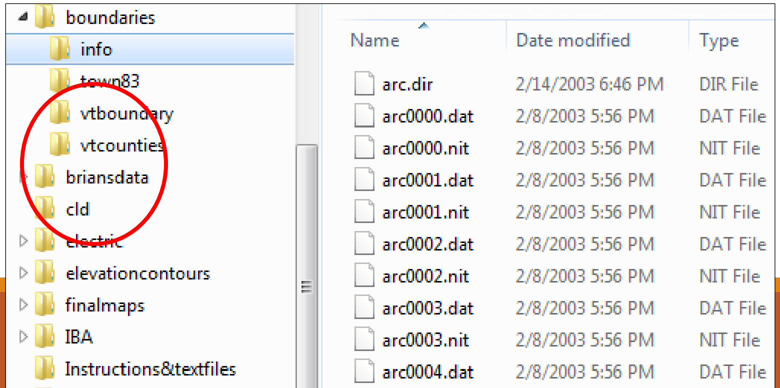
Windows Explorer (file system) views:

Data types

- **Coverage**: native file format for old ArcInfo 7.x (not a file, but a complex directory structure)







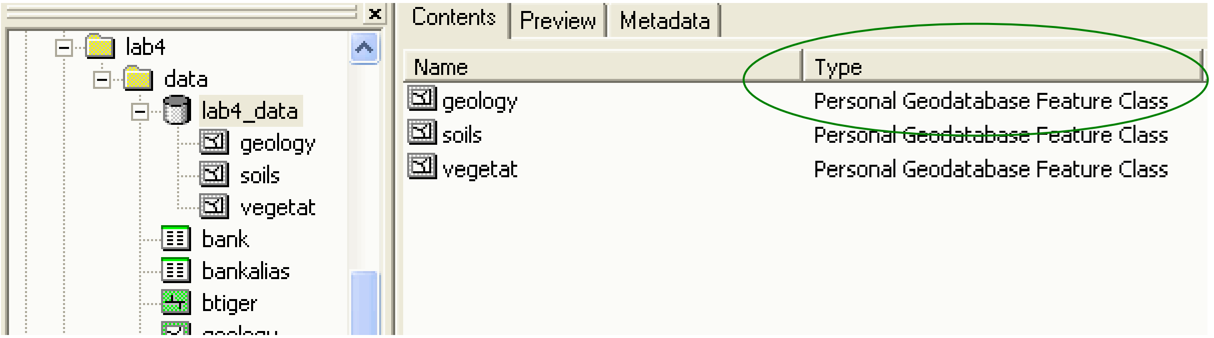
Copy/move/rename the coverage folder in Windows Explorer and you corrupt the whole workspace! (Use ArcCatalog to manage spatial data layers)

Data types

- Tabular data: data table (often non-spatial) that can be used in a GIS
  - Spreadsheet
  - Dbase database file -- .dbf
- Map document: a project file—it stores your map layout and preferences in ArcGIS….but not the data!

Data types

- **Geodatabase** (GDB): This is ArcGIS's newest data model for storing geographic information
  - A geodatabase can contain many layers, known as "feature classes." Shapefiles have only one layer.
  - As an example, here is a geodatabase with three feature classes, as seen in Arc Catalog



Feature classes

- In a geodatabase there are features classes for points, lines and areas (polygons) as well as for rasters (surfaces), annotation (labels), and dimensions

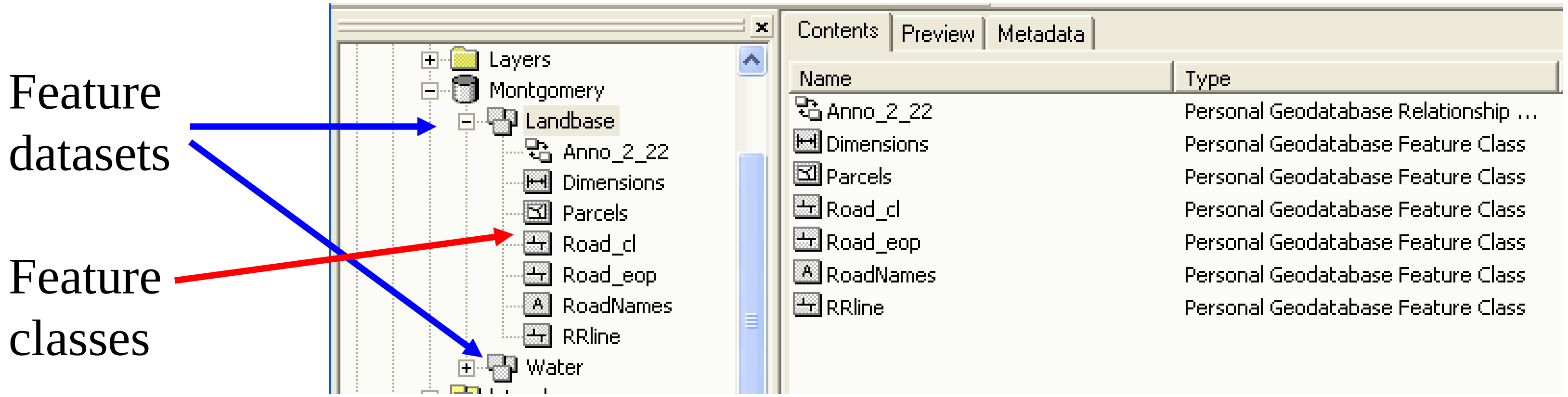
Polygon, or area feature class

Line, or arc feature class

Annotation class: stores info about displaying labels on a map layer; important for roads and utilities

Feature dataset

- Feature dataset: a thematic grouping of feature classes
- Feature classes can be stand alone (do not have to be within a feature dataset)



Geodatabase

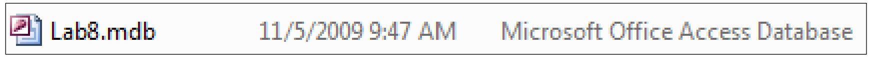
- Geodatabases offer numerous advantages:
  - Multiple spatial layers and non-spatial data sources can be stored in a single file (personal GDB) and organized thematically
  - Rules can be easily defined that can apply to all classes
  - These rules can include relationships between layers
  - Example: one subclass with water lines and one with water valves; you can ensure that if you move a water line, the water valve that connects that line will move as well, or it ensures that, say, if the material attribute for a water line is set to copper, the water valves that connect to it will also be copper

Geodatabase

- Other advantages:
  - Spatial layers maintain the same measurement and geographic reference systems, because new layers added to the geodatabase can “inherit” properties of existing ones
  - Labeling “behavior” can be stored as an annotation feature class, making it easier to keep labels consistent
  - “Domains” can be established, which specify the valid ranges of values for attributes, which reduces input error.

Geodatabase

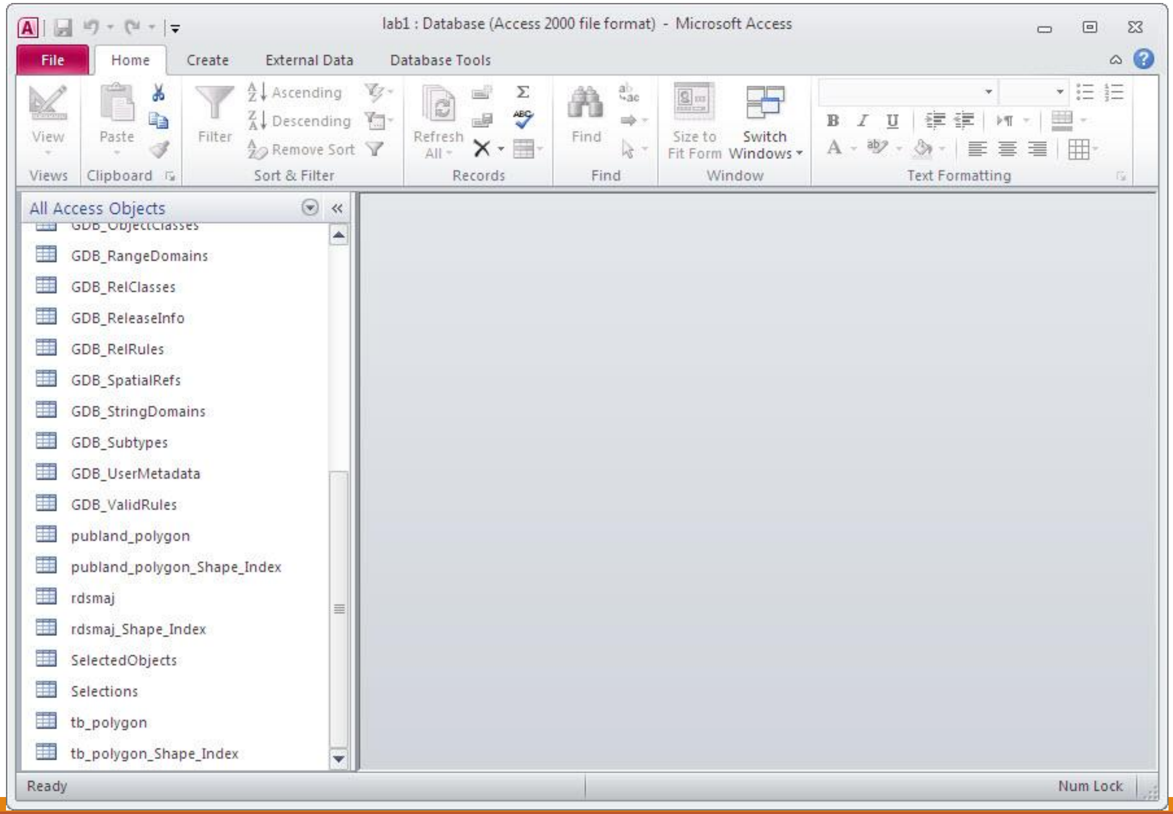
- Types: Personal, File or Enterprise (multi-user)
  - We will use personal geodatabases. These are actually Microsoft Access files with the .MDB file extension.



  - We may also use the newer *file* geodatabases. These are a multi-file-within-folder model similar to a *coverage*, but easier to manage. The folder has a .GDB extension.
  - A multi-user (SDE) geodatabase is for organizations that wish to have many people accessing the same database. These are used with enterprise database systems, like Oracle, Informix or SQL Server

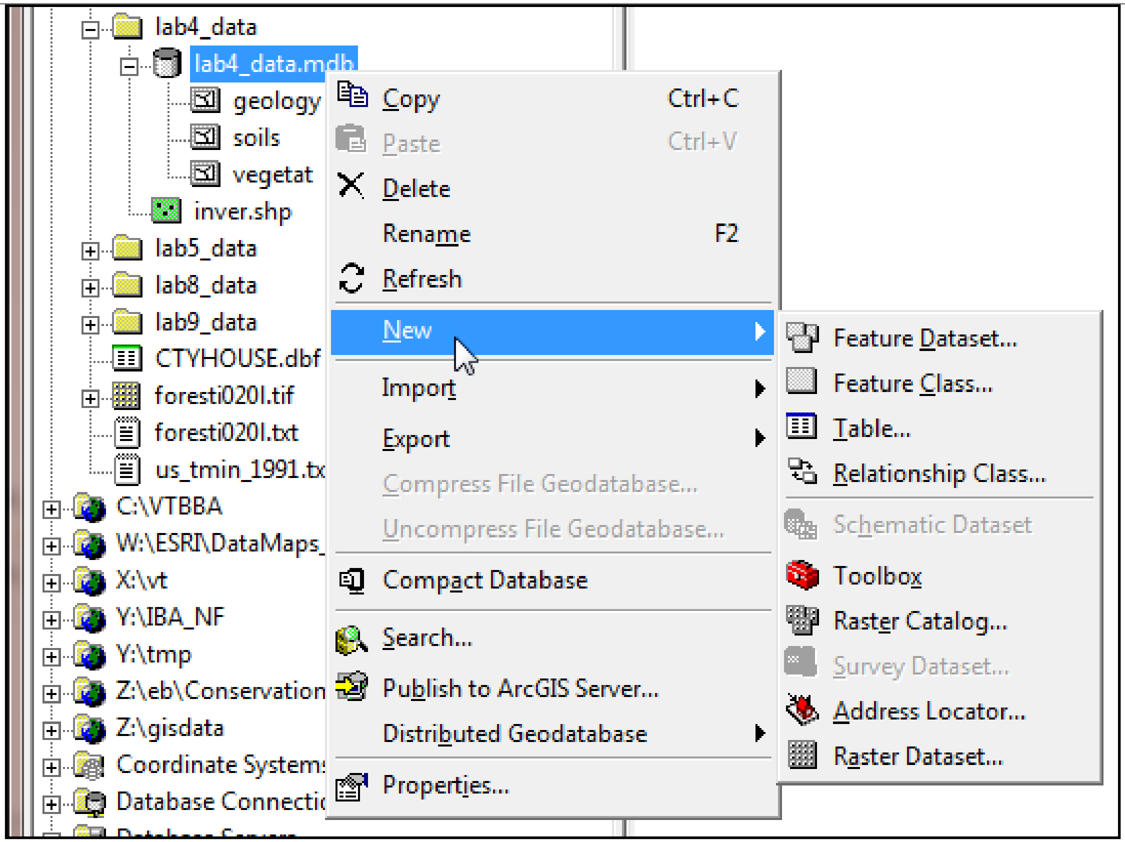
Personal Geodatabase

- View of a geodatabase in MS Access, where you can edit and query attribute tables and manage relationships between feature classes….
- Can only view spatially in ArcGIS (not in Access)



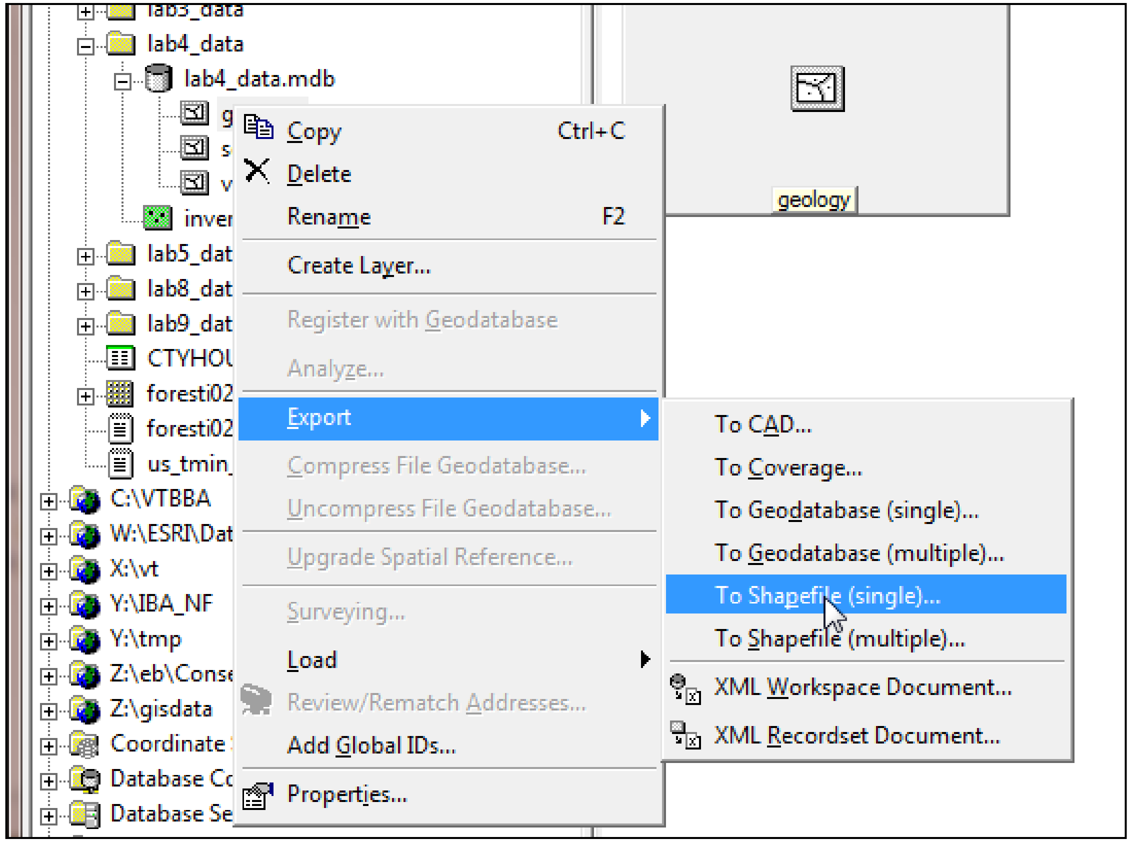
ArcCatalog: Data management

- Create and modify geodatabases.
- You can create new feature classes, tables or relationship classes within a geodatabase



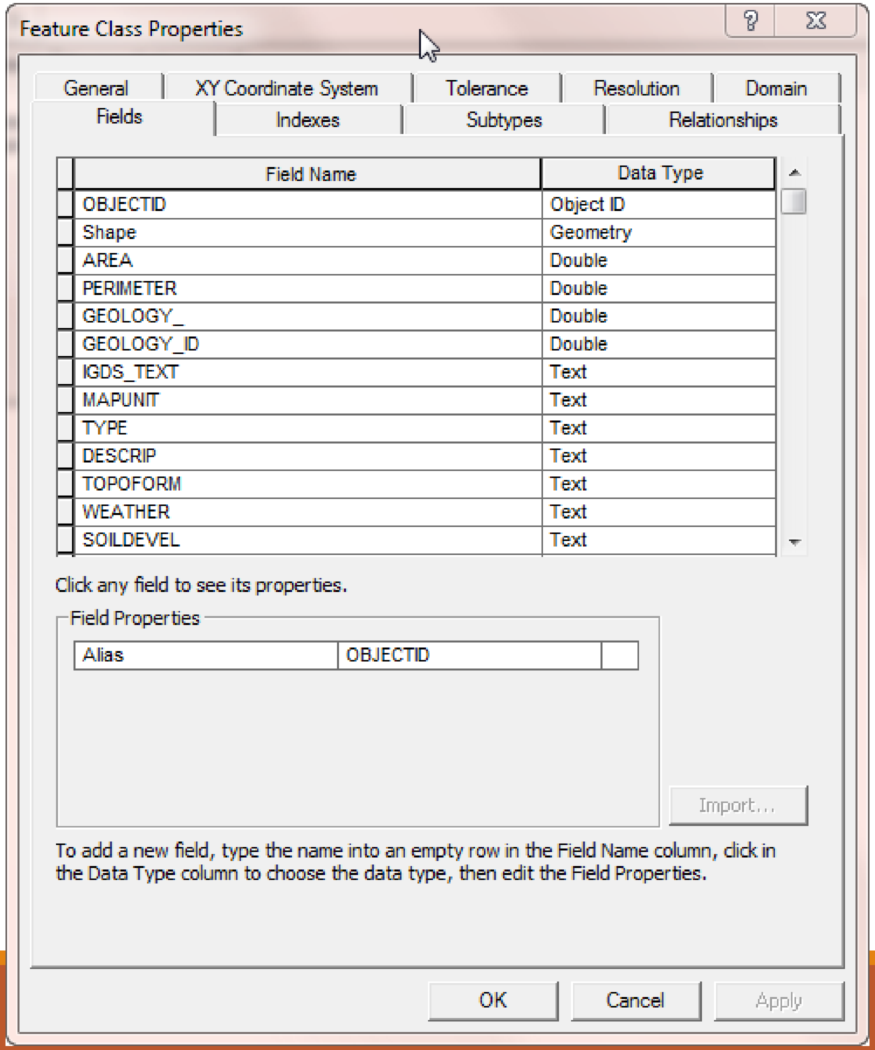
ArcCatalog: Data management

You can also import existing shapefiles or coverages into a geodatabase or export a feature class to a shapefile



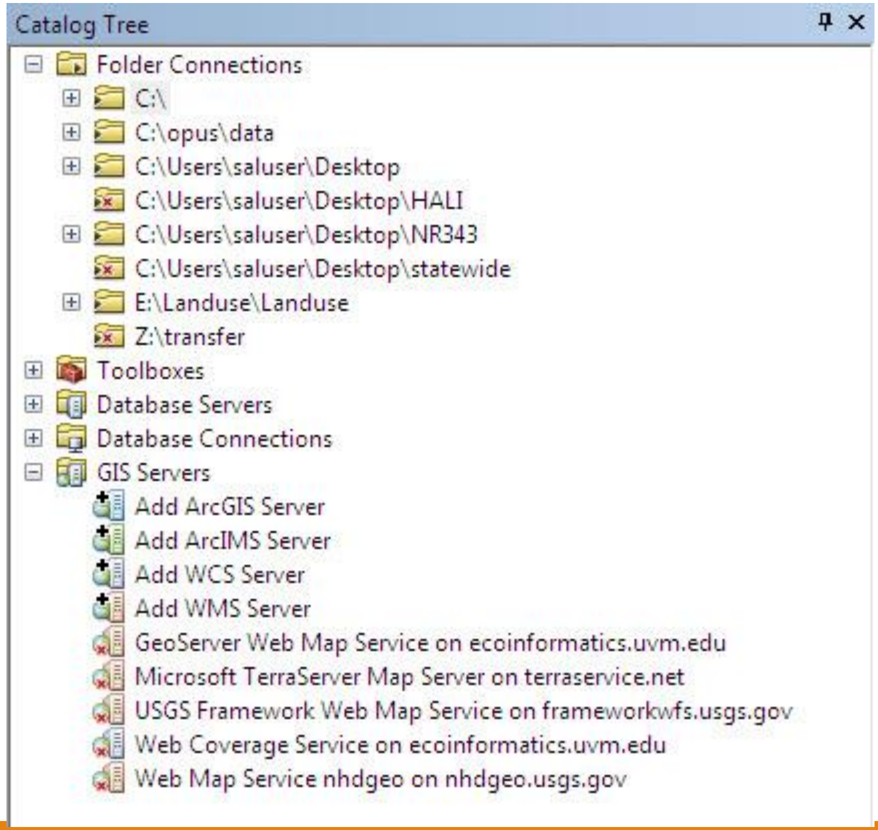
ArcCatalog: Data management

Through right clicking, you can access the properties of a feature class and make changes … e.g., alter fields for the feature class' attribute table



ArcCatalog: Online data

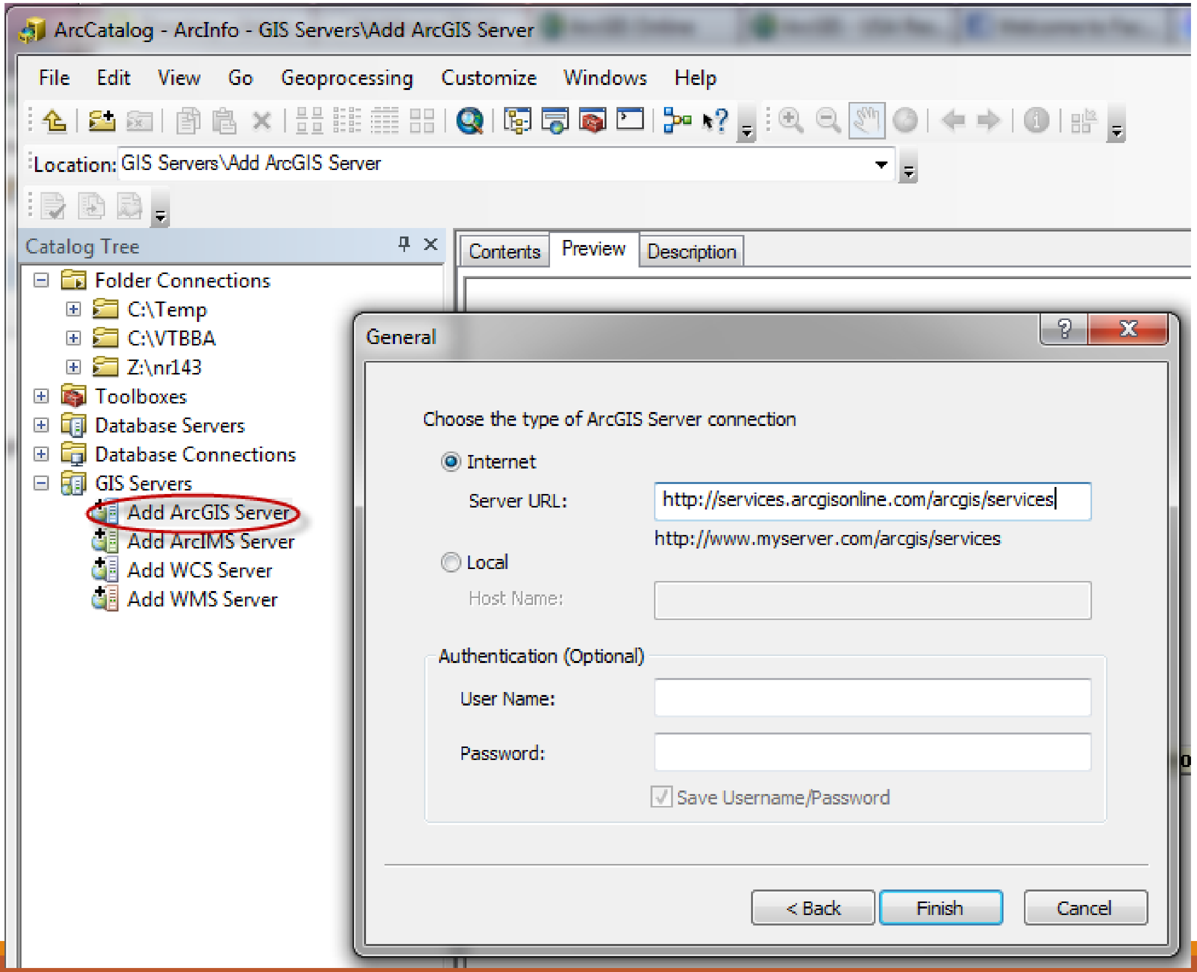
- ArcCatalog also allows you to access data directly from the Internet



ArcCatalog: Online data

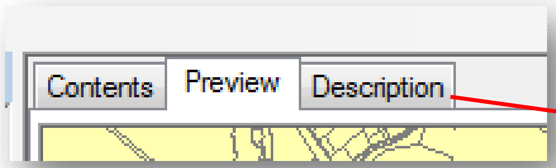
ArcGIS Online

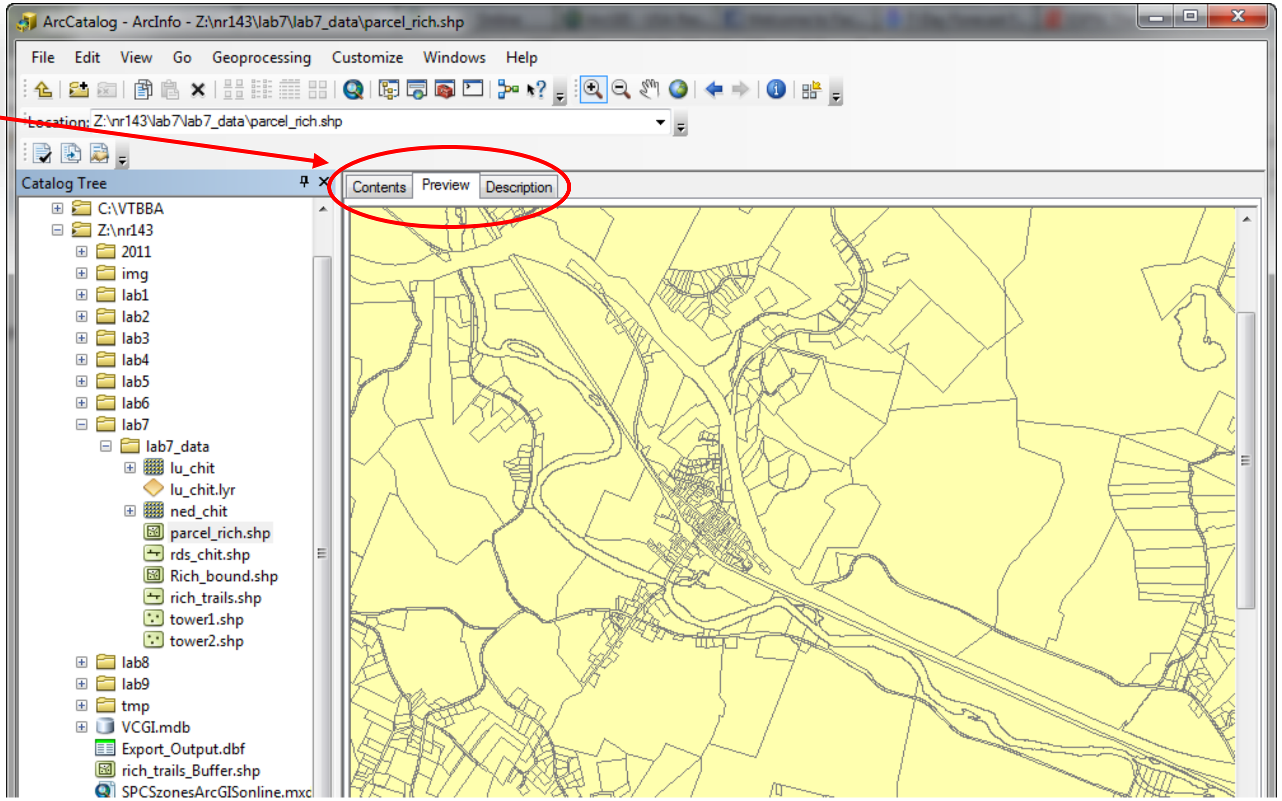
(arcgis.com)



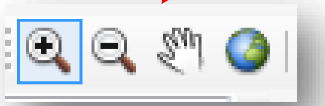
ArcCatalog: Functionality

- ArcCatalog allows you to preview geographic data, including geodatabase feature classes



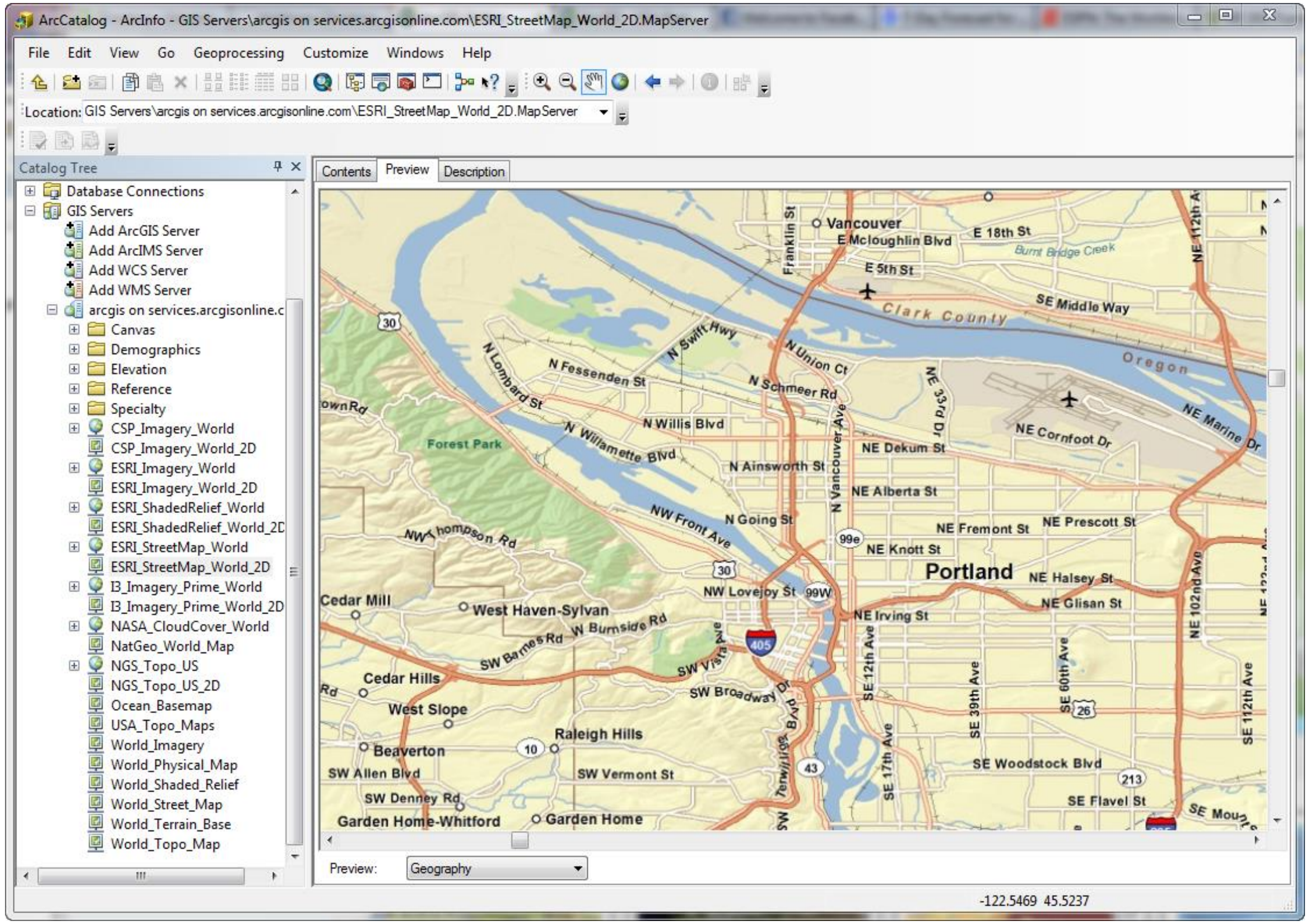


Use these buttons to zoom and pan



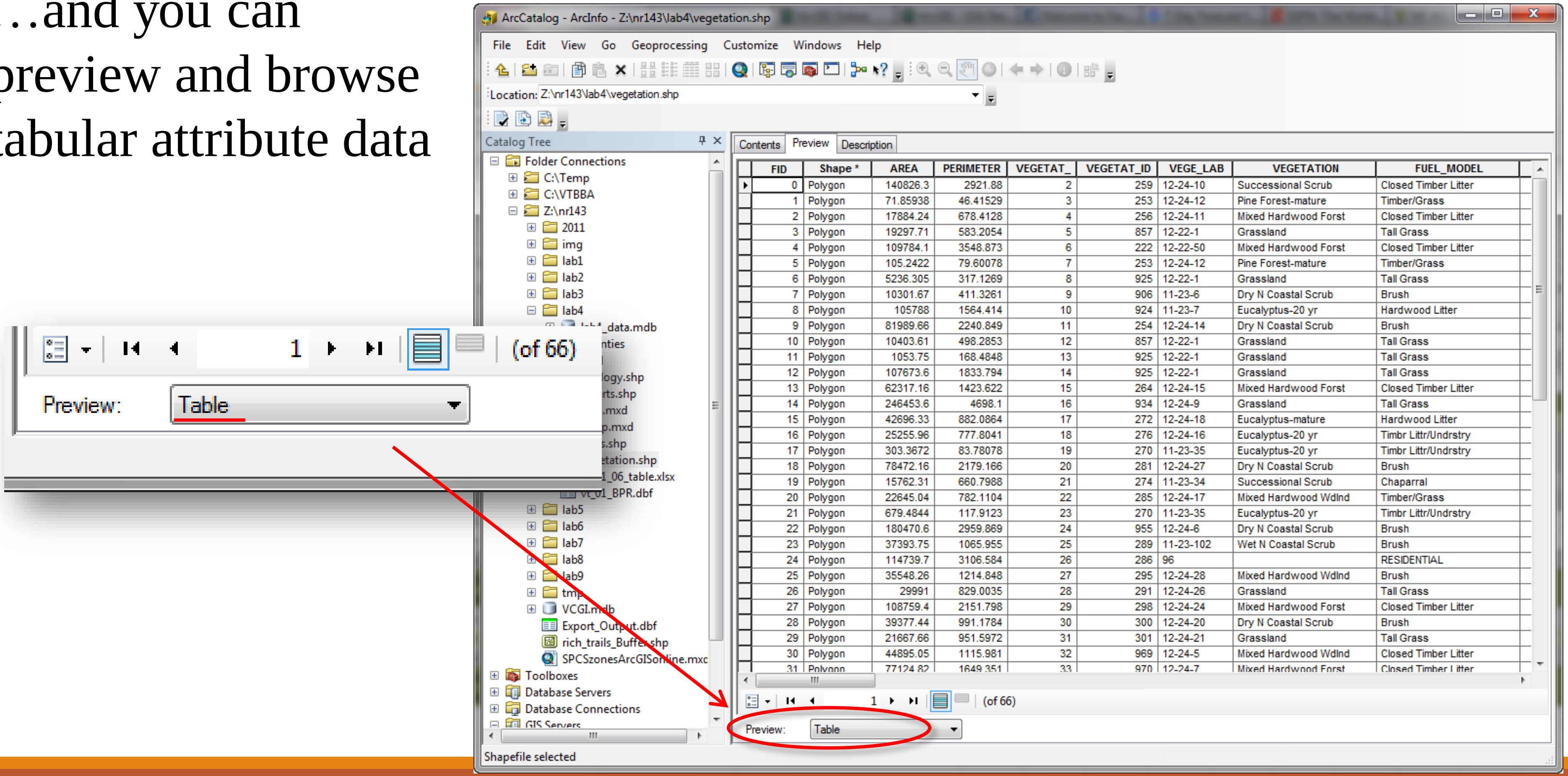
ArcCatalog: Functionality

- You can also preview online data sets



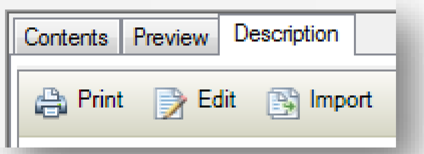
ArcCatalog: Functionality

…and you can preview and browse tabular attribute data

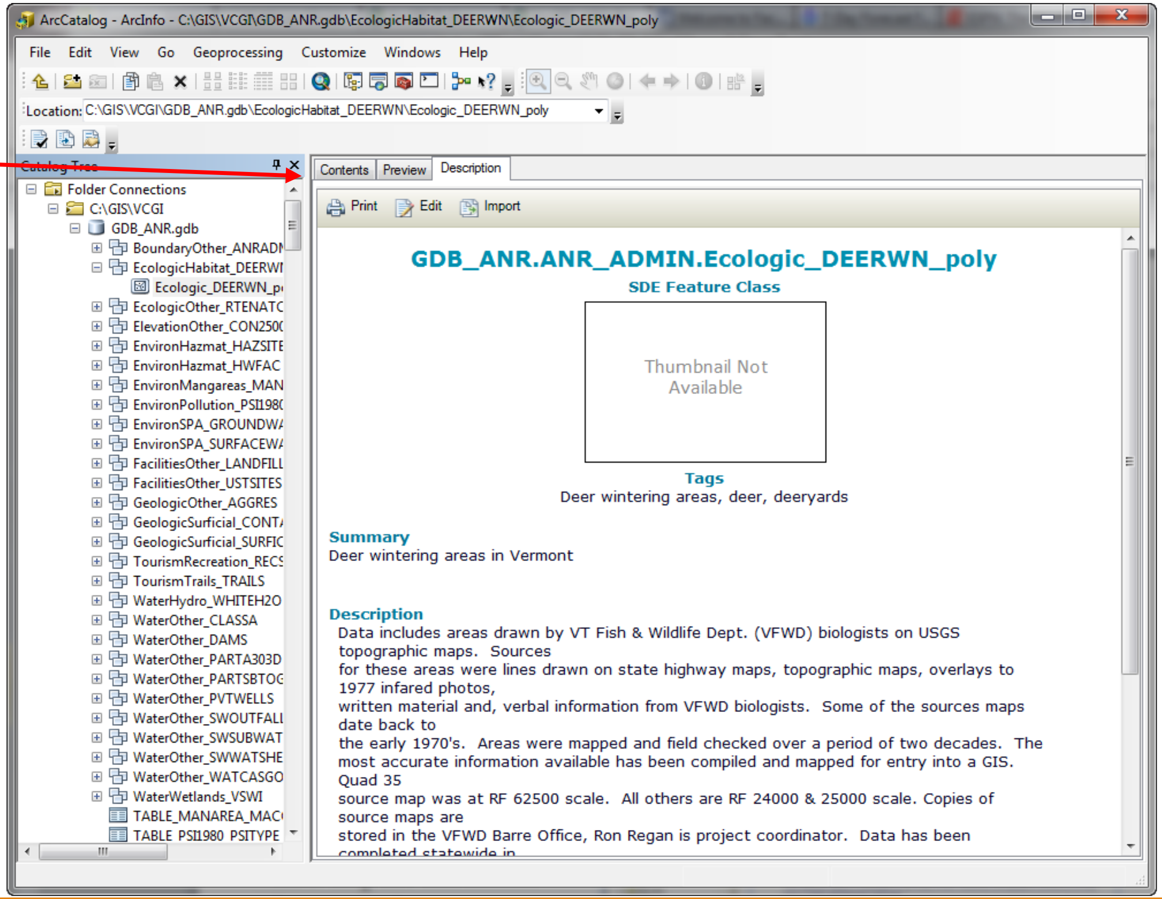


ArcCatalog: Functionality

- Preview, create and modify the “metadata,” or data about the data

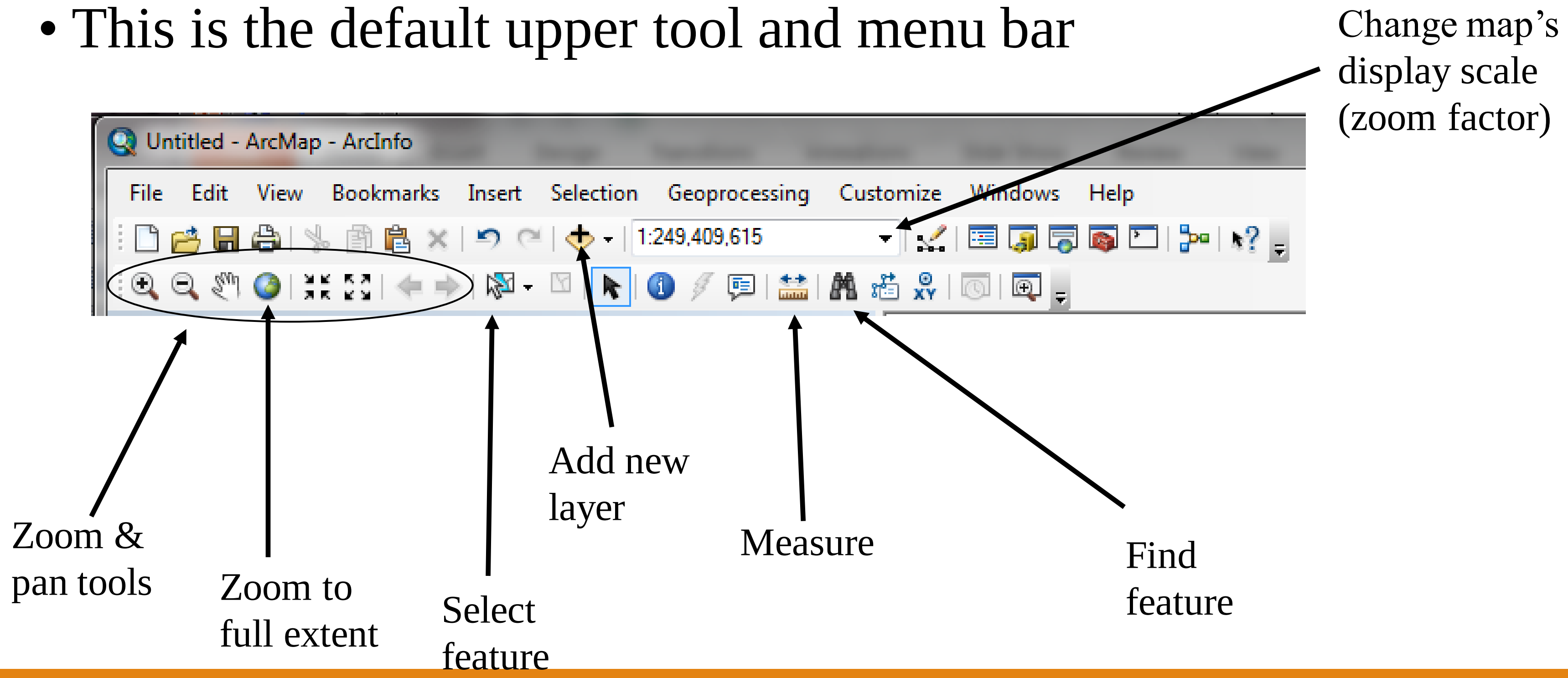


ArcCatalog standardizes this information and allows you to import and export metadata between layers



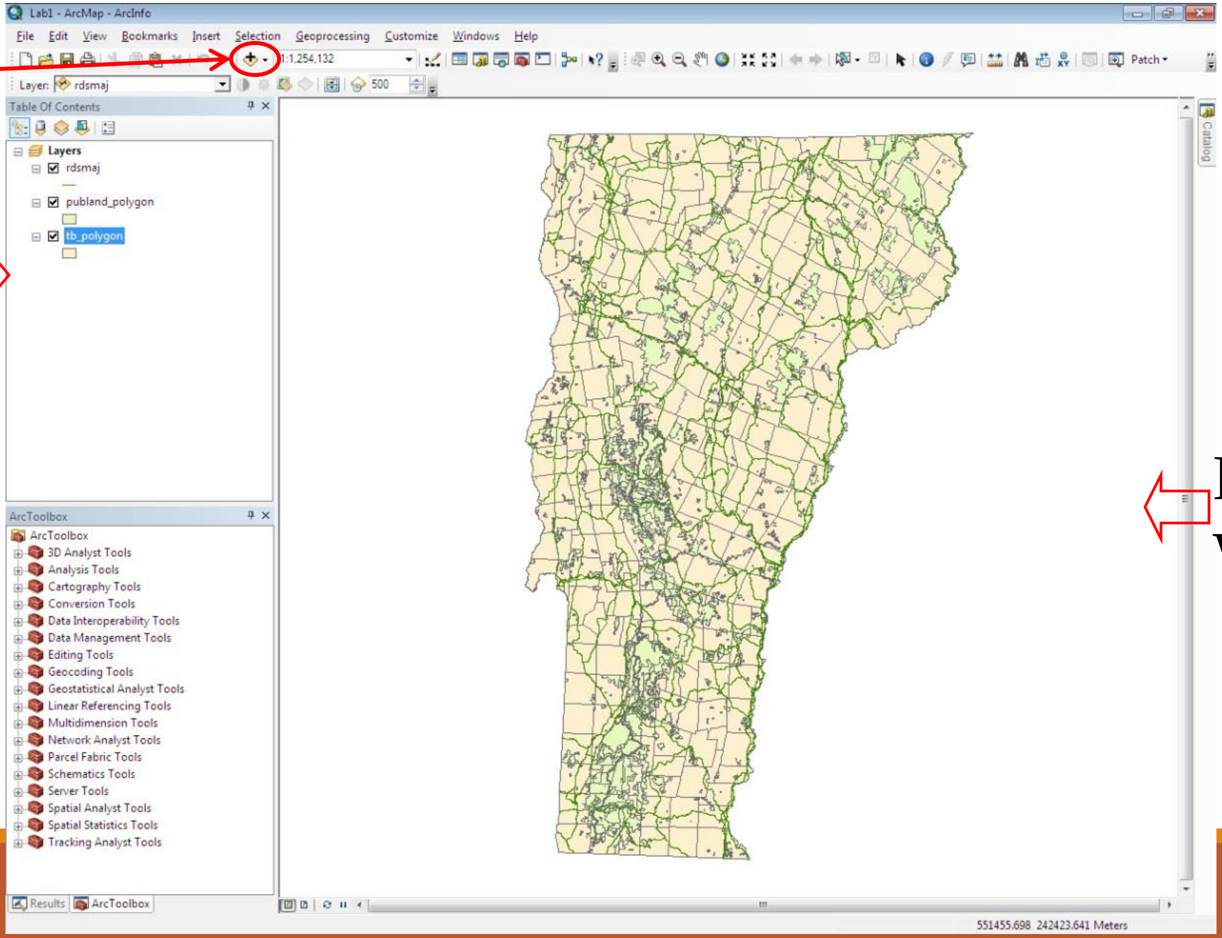
ArcMap

- ArcMap is where you'll do most of your mapping and visualization of spatial data
- This is the default upper tool and menu bar



ArcMap: overlaying data

• Using the add data button, you can overlay any number of layers for which geographic reference information exists.

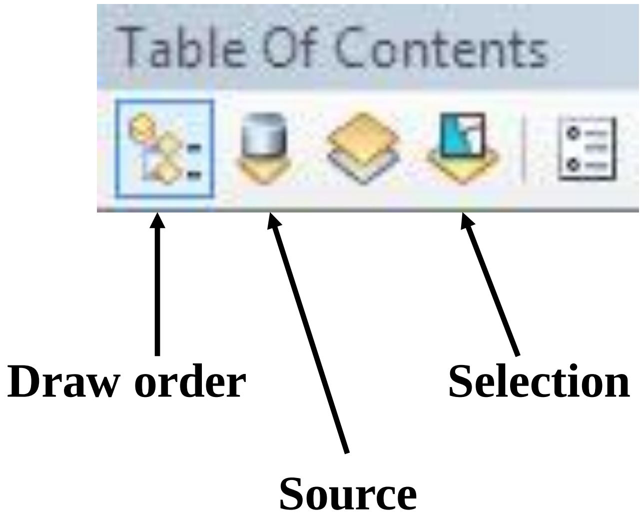


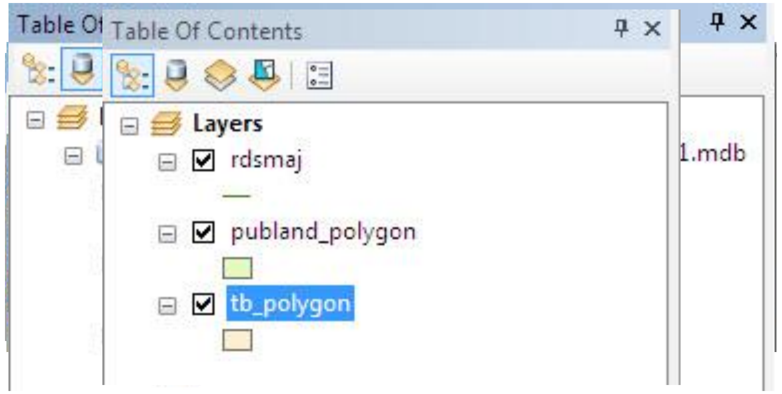
TOC

Display Window

ArcMap: Table of Contents

- The window on the left side of ArcMap is the Table of Contents, where project data layers are listed
- There are several TOC views, including display and source

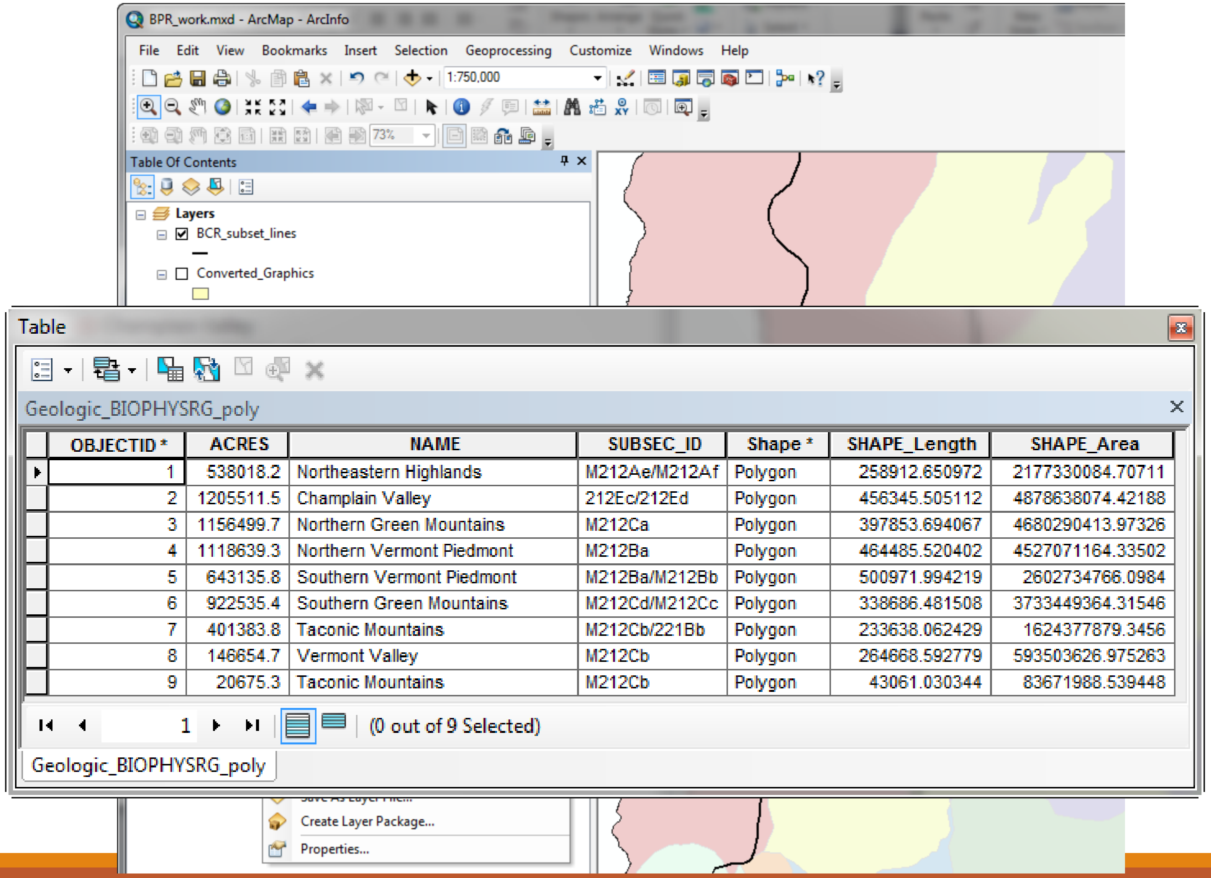




ArcMap: properties

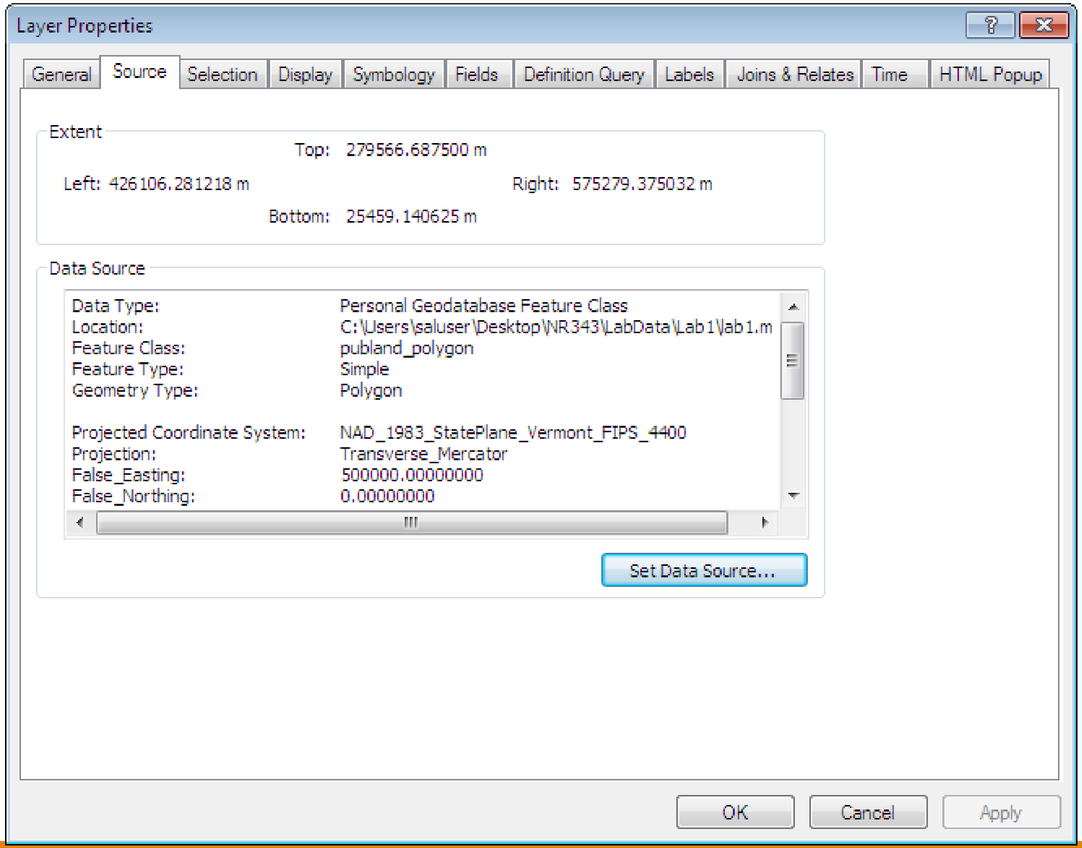
- You can access many functions by right clicking on the layer in the TOC.

You can access, edit and query attribute tables from a special interface



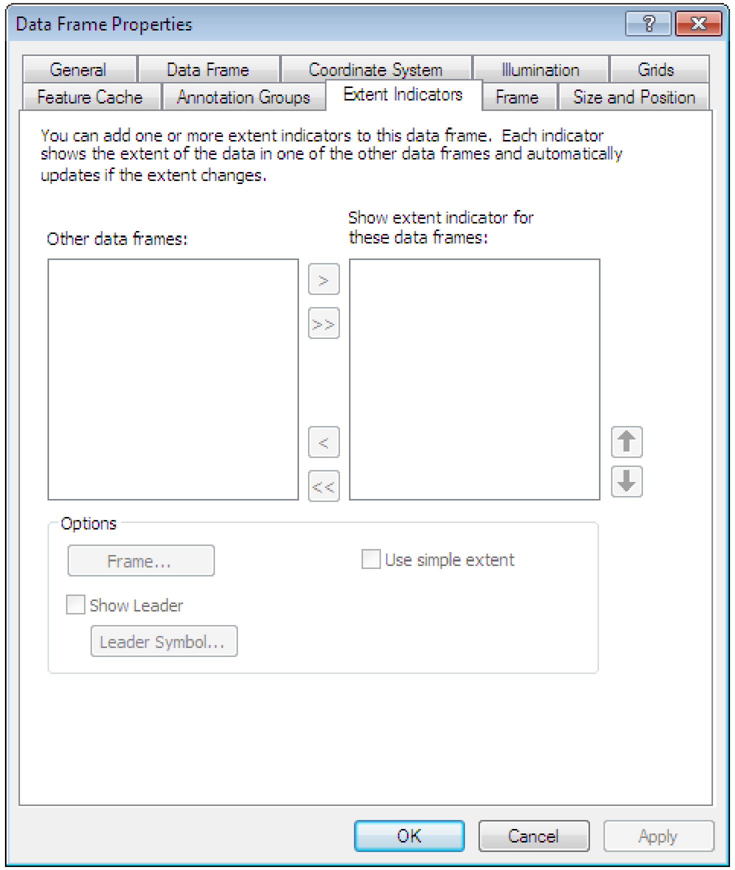
ArcMap: Layer properties

- You can access layer properties by double clicking on that layer: much functionality is found there



ArcMap: Data frame

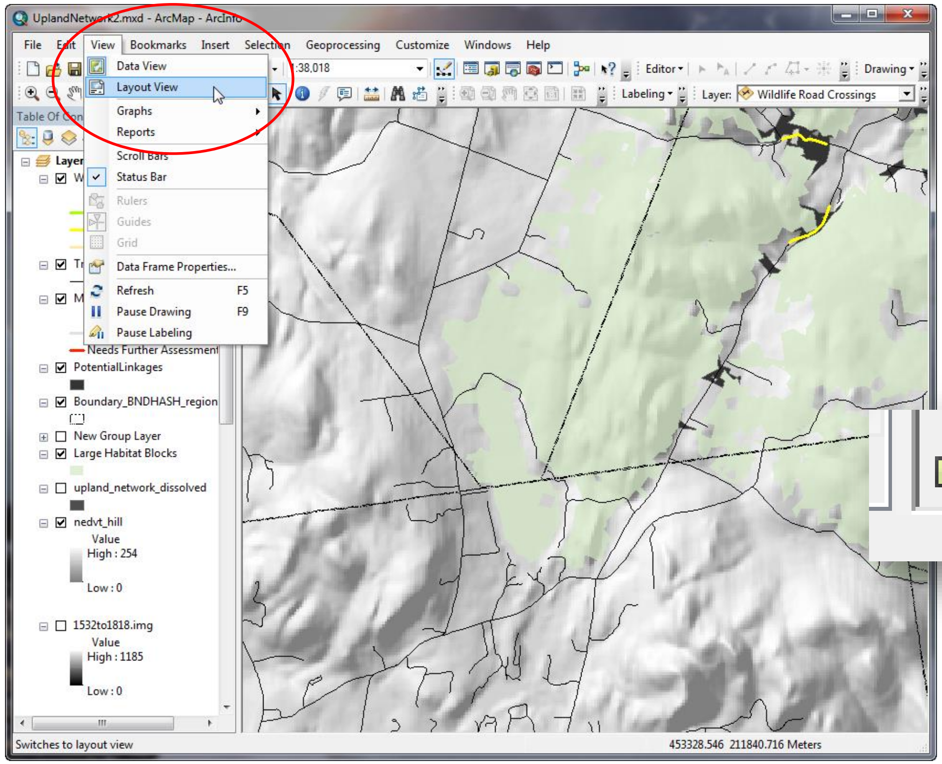
You can access properties for the whole data frame by double clicking on “Layers” at the top of the TOC or by right clicking on the data frame



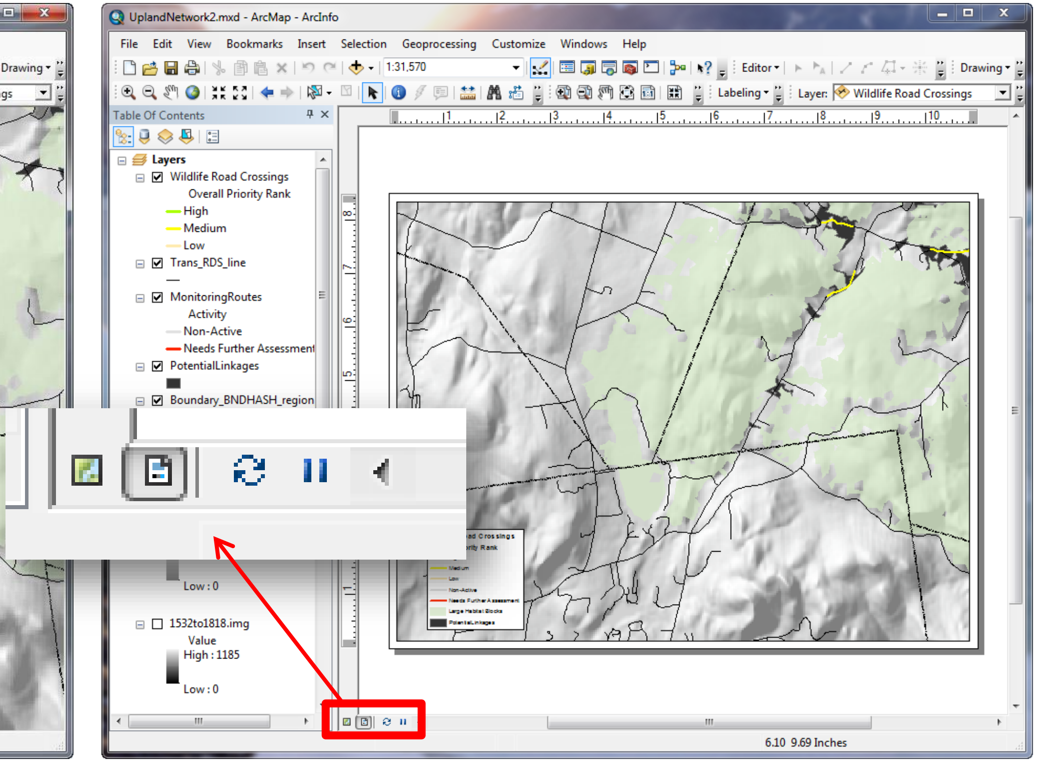
ArcMap: Content Pane

- ArcMap can view in two modes:

Data view: for viewing, analyzing and manipulating data

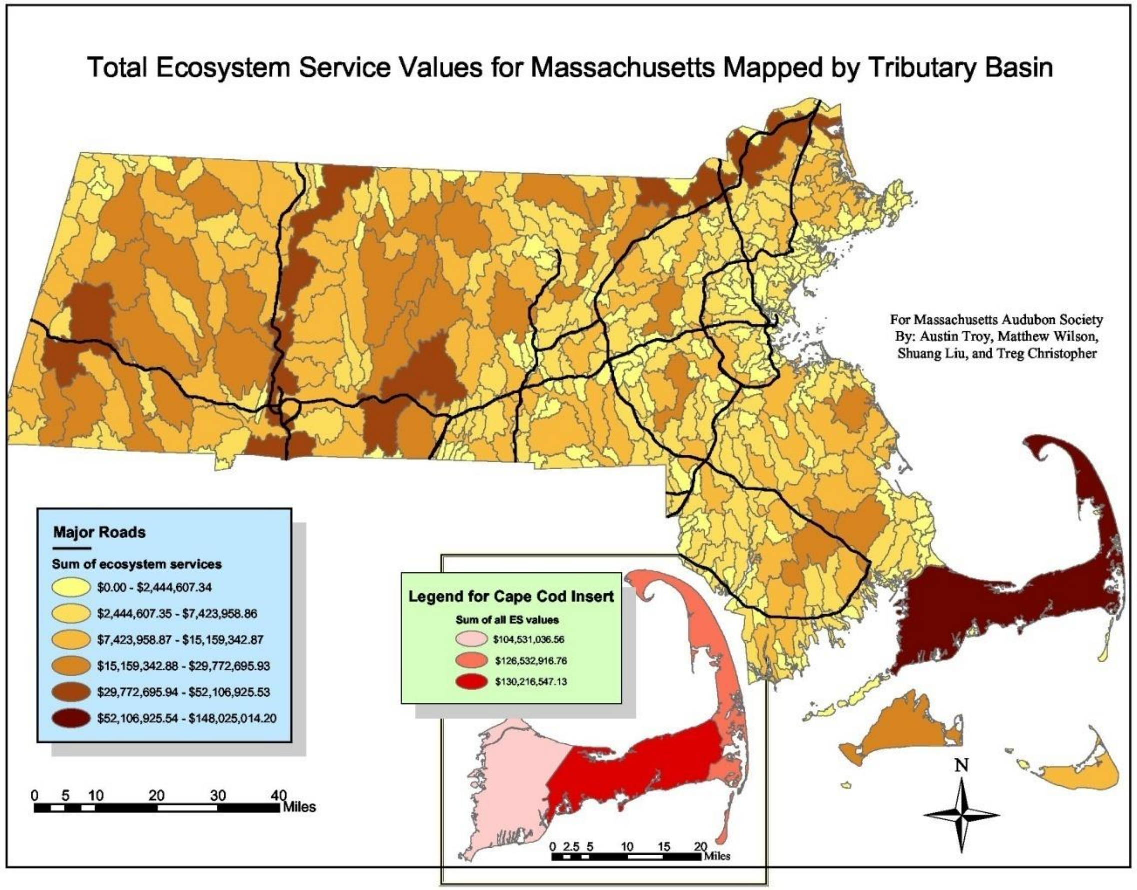


Layout view: for laying out data for presentation



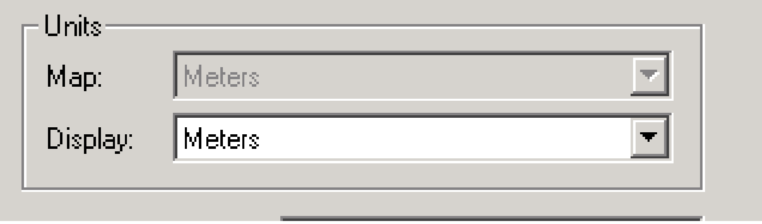
ArcMap: Layouts

Maps are created in the layout view, where titles, legends, north arrows, scale bars, and other elements can be added and arranged.

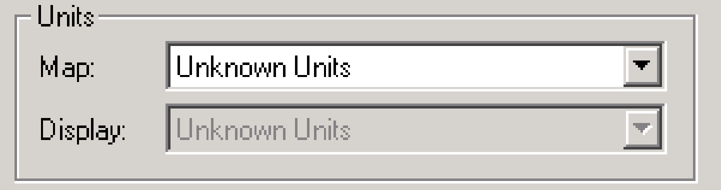


ArcMap: Data frame—Units

•Two of the most important properties are Map and Display units. Where the Map units are already set (because reference info exists) they will be "grayed out" but display units are still changeable.



When the units are unknown to the system (reference info not embedded in the file) you'll see:

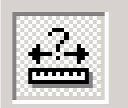


This turns into guesswork

ArcMap: Units

• Once you've set display units, future geographic measurement functions should be in those units, even if they're different from the Map units; however, this does not apply to non-spatial attribute values

• If you use the ruler tool for instance, measurements will likely be in the selected Display units at the bottom of the page unless you've selected different units in the ruler tool interface



ArcMap: Scale

- Translation factor between one unit on the map and same unit in the real world
    - map units: ground units (ratio)
    - 1:10,000 means that 1 cm equals 10,000 cms in real world

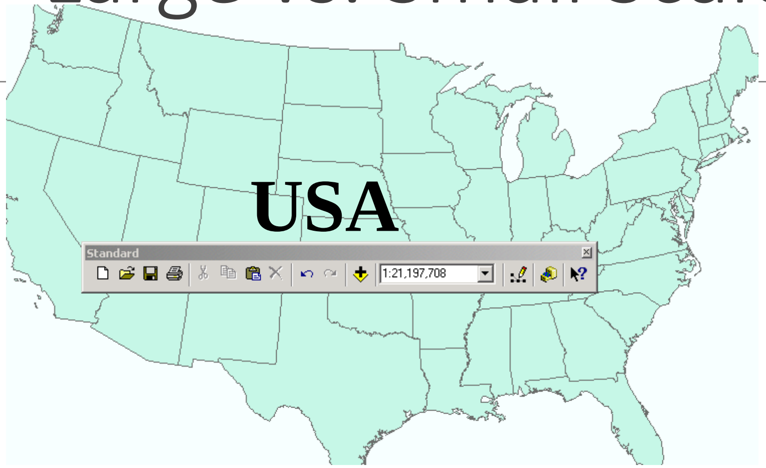
1cm is 1.98 million cm in real world

- Scale is a <u>dimensionless number</u>

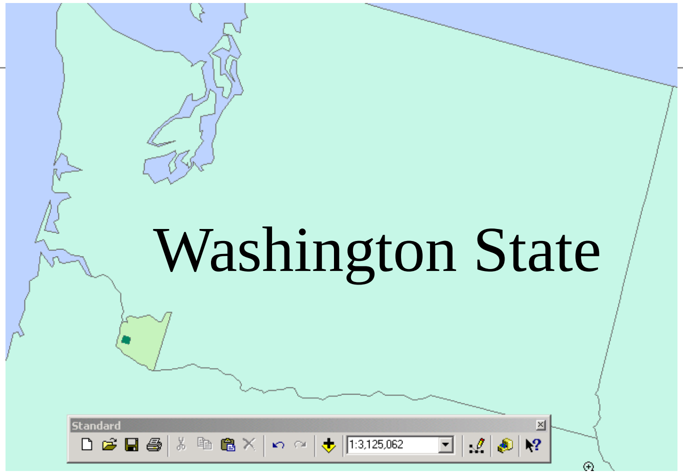
Large vs. Small Scale

- Large scale
  - Lots of detail
  - Shows small features
- Small scale
  - No much detail
  - Shows large features

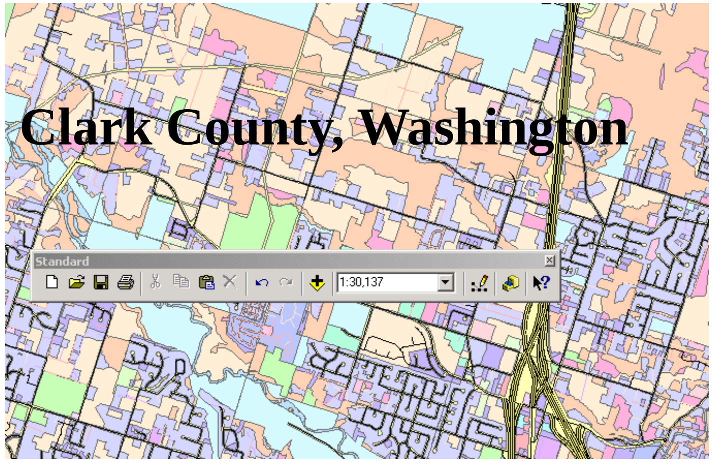
Large vs. Small Scale



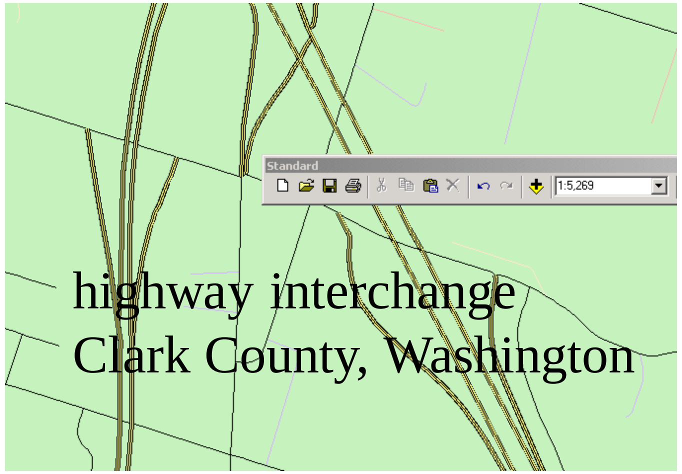
1:21,197,708



1:3,125,078



1:30,137



1:5,269

ArcGIS: Extensions

- Specialized applications

  - 3D Analyst

  - Spatial Analyst

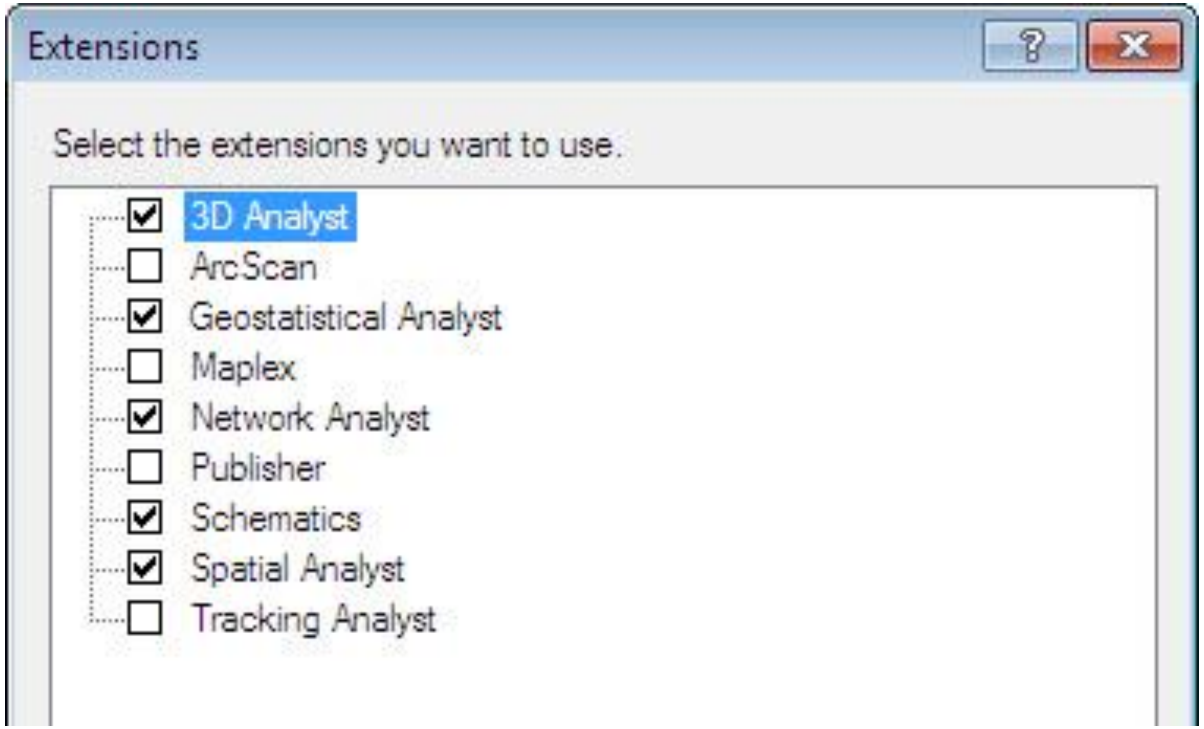
  - Geostatistical Analyst

  - Tracking Analyst, X Tools…

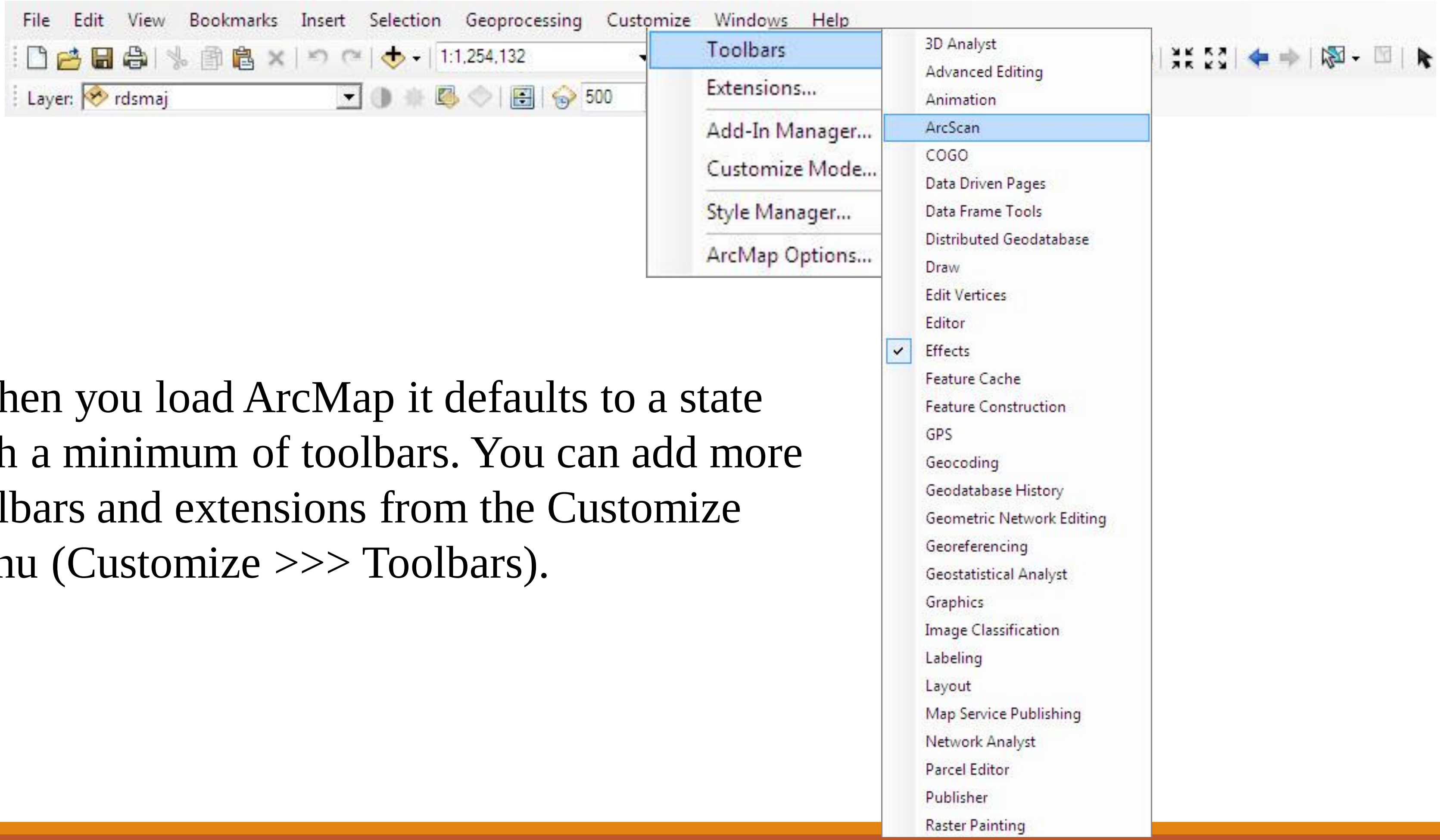
  - Others…

ArcGIS: Extensions

- Activate by going to Customize>>extensions



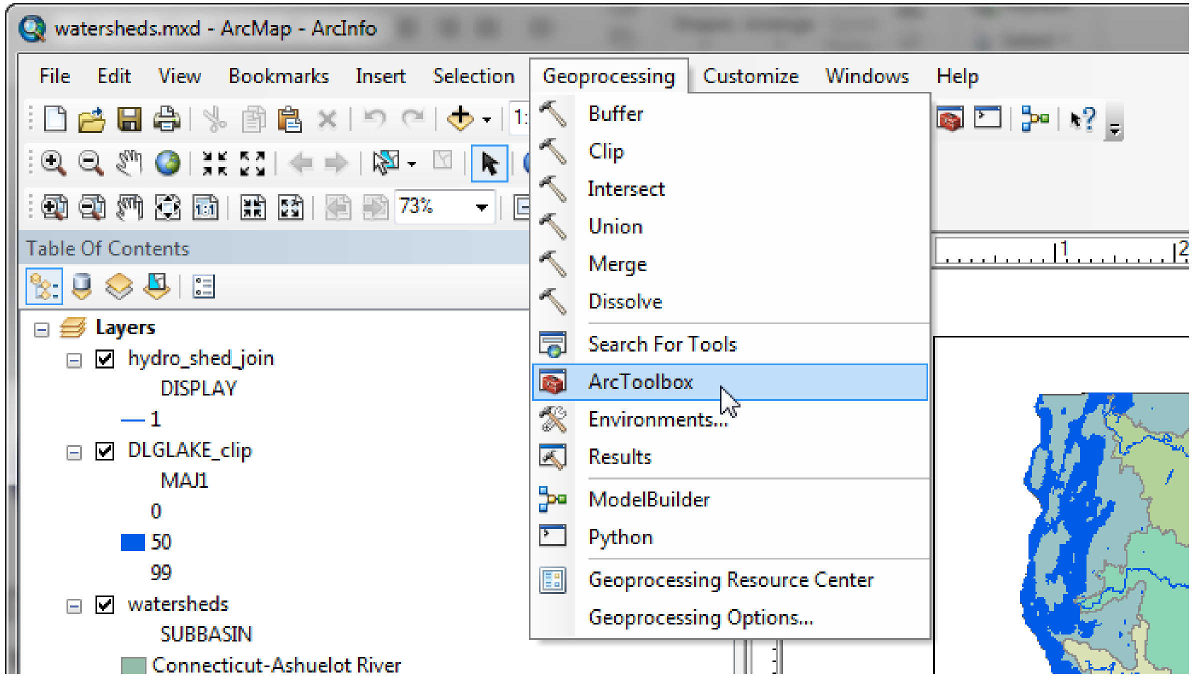
ArcMap: Toolbars

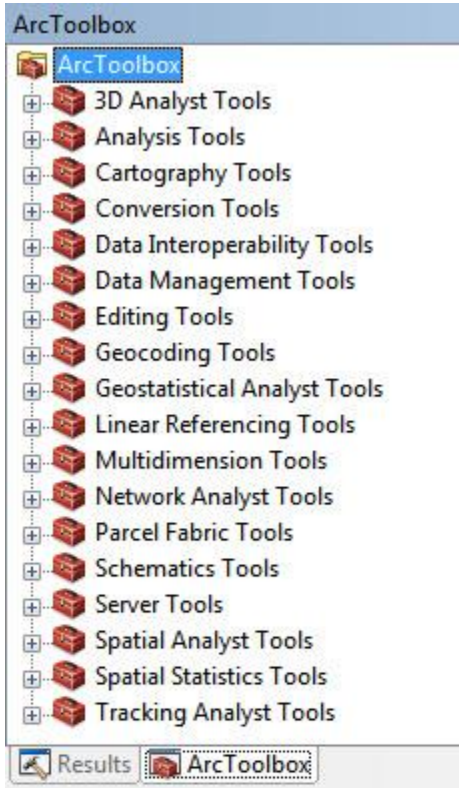


- When you load ArcMap it defaults to a state with a minimum of toolbars. You can add more toolbars and extensions from the Customize menu (Customize >>> Toolbars).

ArcToolbox

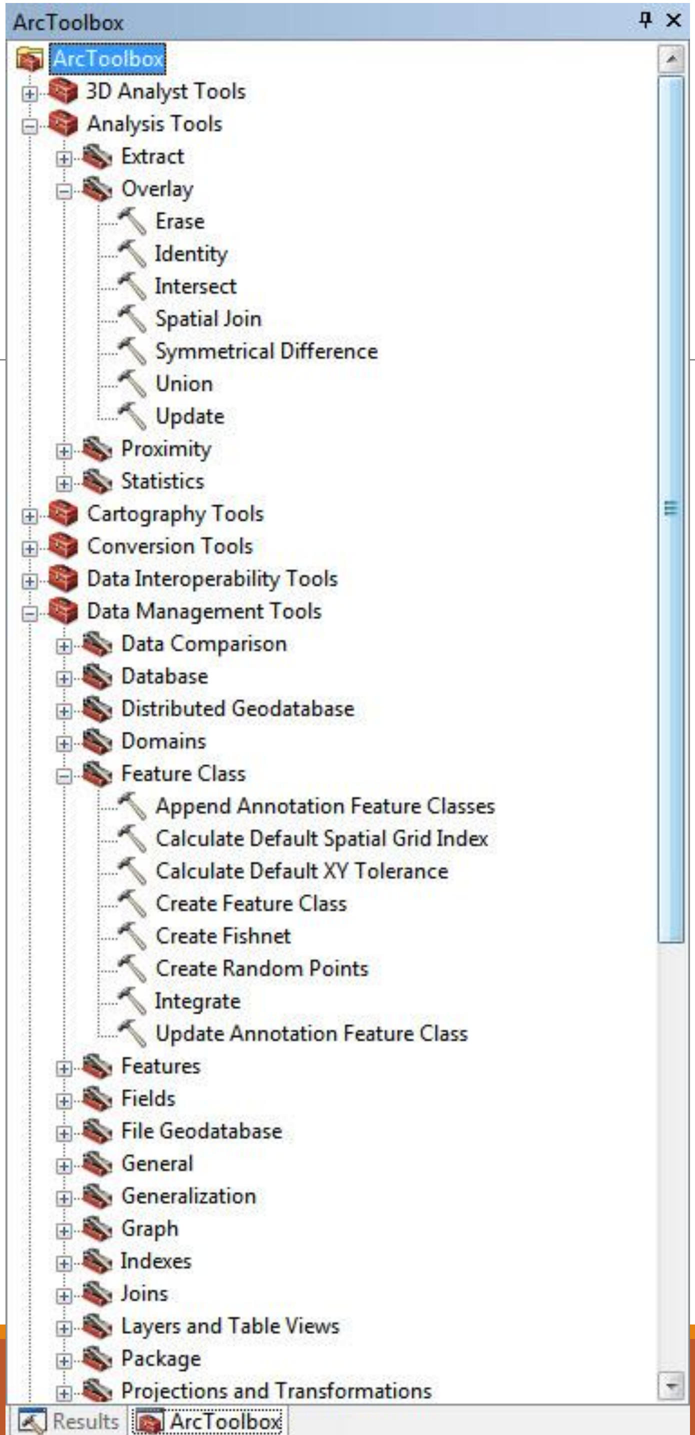
- Embedded within both ArcMap & ArcCatalog





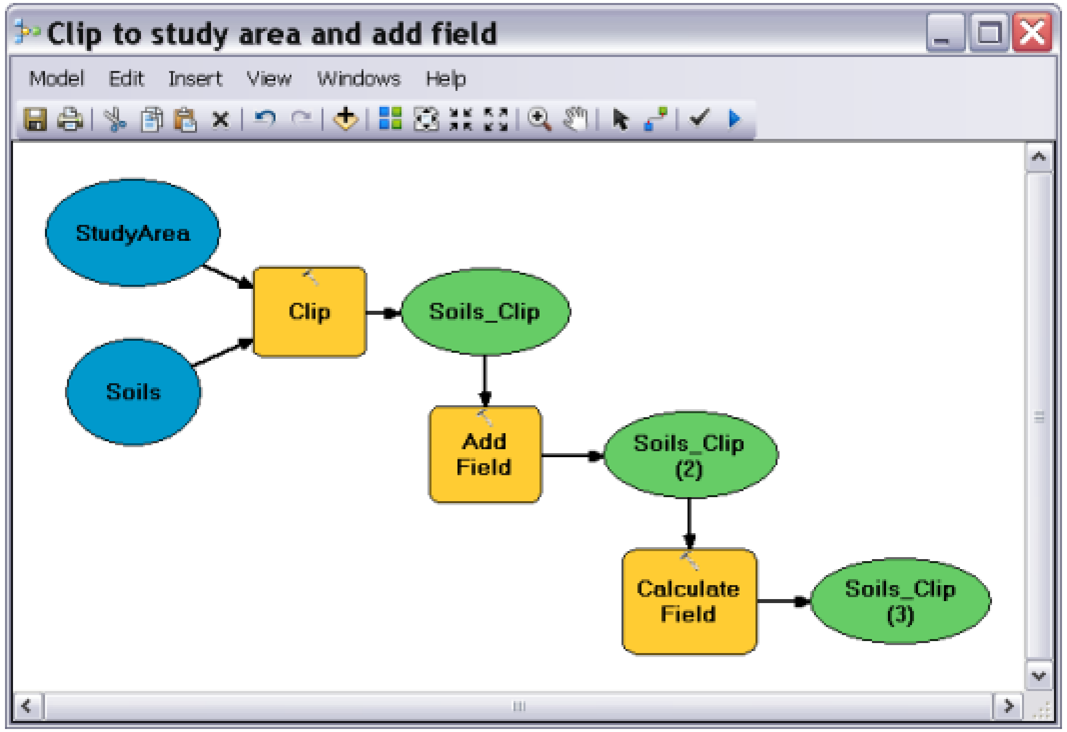
ArcToolbox

- Advanced Analyses
  - Extract
  - Overlay
  - Proximity
  - Statistics
- Data Management
- Conversion Tools
- Etc.



ModelBuilder

- An application for building models – workflows that string together sequences of tools



ArcScene

Primarily for viewing 3D data